

The westward transmission of Indus Valley sealing technology: origin and development of the ‘Gulf Type’ seal and other administrative technologies in Early Dilmun, c.2100–2000 BC

This paper provides a comprehensive examination of the seals of the so-called ‘Gulf Type’, which date to the end of the third millennium BC. It is argued that the Gulf Type seals are of key importance to our understanding of the origin of sealing and other administrative technologies within an emerging Early Dilmun ‘state’. Based on principal component analysis it is demonstrated that the shape of Gulf Type seals with inscriptions in Indus characters is distinct from seals without inscriptions. It is further argued that Gulf Type seals found in the Indus Valley, Iran, Mesopotamia/Bahrain and Bahrain can be connected with relatively discrete morphological groups apparently corresponding to different areas of production. The Indus inscriptions on the seals are investigated with particular emphasis on the abnormal occurrence of prefixed ‘twins’ signs in the western inscriptions. The hypothesis that a language different from that of the Harappans was used on these seals is reconfirmed on the basis of a newly found seal with a particular instructive pseudo-inscription. The paper concludes that break-away Harappans operating in the western orbit invented the Gulf Type seals but that the type from around 2050 BC became practically synonymous with the merchant communities in Dilmun.

Keywords: technology, Dilmun, Bahrain, Gulf Type seals, Indus Valley, trade, Ur III

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Introduction¹

The occurrence in Mesopotamia of conventional square Harappan seals with a heraldic animal surmounted by an inscription in Indus characters is generally thought to mirror the actual presence of Indus Valley merchants in Babylonia (Mackay 1925; Langdon 1931; Gadd 1932; Collon 1994;

Parpola 1994a). Conversely, ever since the first discovery of circular seals with Indus text in various locations in Mesopotamia and Bahrain, commentators have been more hesitant with regard to the cultural affinity of this nonconforming type, which has only rarely been encountered in the Indus Valley. The occasional presence of Indus text on the early circular seals has puzzled researchers and focused attention on the potential of this class of objects to broaden our understanding of the relationship between the Indus Valley and

¹ I would like to thank Flemming Højlund, Moesgaard Museum, Denmark, for reading and commenting on earlier versions of this manuscript.

Mesopotamia. In this respect too little attention has been paid to the important fact that these early seals embody the transmission of vital administrative technology into the Dilmun culture simultaneously with the rise of an early 'state' on Bahrain (Højlund 2007: 124–125).

On a superficial level these circular seals with Indus inscriptions are, except for the presence of text, often indistinguishable from the seals of so-called Arabian (Persian) Gulf Type without inscription, which have been found in very substantial numbers particularly on Bahrain Island (Kjærøum 1994; 2007; Al-Sindi 1999).

By far the most abundant seals are of the so-called Dilmun Type, which is distinguished from all others by consistently having three grooves and four dots-in-circles on the reverse. The reported number of circular 'pre-Dilmun Type' stamp seals has however also seen a considerable increase over the years. So far only moderate attempts have been made to penetrate the apparent uniformity and break up the class of early 'Gulf' seals into culturally significant variants. In the present study an effort to do this has been made to increase the knowledge of the spread of sealing technology, and by implication possibly also of writing and weight standards into the Dilmun culture.

Even if the production of circular seals of pre-Dilmun types has been linked to workshops on Bahrain Island (Glob 1959: 338; Kjærøum 1994: 338, fig. 1753; During Caspers 1977), we are left with a list of essential but poorly answered questions concerning the situation surrounding the adaptation of Dilmun's first indigenous sealing technology from the Indus Valley:

- By what route did this sealing technology spread?
- Who were the agents instrumental in its transmission?
- When did it happen and what role did it play in the emergence of social complexity in Early Dilmun society?

The study at hand is designed to improve the basis for addressing these fundamental questions, by a comprehensive re-examination of the earliest corpus of seals. Particular emphasis is placed on the seals with Indus texts as they represent an essential source of understanding of the westward transmission of sealing and other administrative

technologies, including conventions for writing and standard weight systems invented in the Harappan sphere.

Previous thoughts concerning the origins of the 'Gulf' seals

Circular seals with Indus text appeared before the academic community in 1932 when Gadd communicated eight such specimens in his important paper 'Seals of ancient Indian style found in Ur' (1932), in addition to a seal presented by Langdon (1932). Based on the evidence that existed at the time, Gadd fully appreciated the significance of the 'new' circular type and advanced the following explanation for their morphological distinction from those of confirmed Indus Valley origin: 'As to the prevalence of the round shape over the rectangular — the direct opposite of what is found in Mohenjo-Daro — this is very marked among the specimens at present recovered from Babylonia, but it is hardly possible to draw any inference from this fact save that the import into Babylonia probably did not come from Mohenjo-Daro or Harappa. But since it is evident that many other sites of the same civilization must have existed, and will doubtless in time be explored, this conclusion is of no great interest' (Gadd 1932: 204–205).

The assumption that these round stamp seals came from an Indus Valley site, other than Mohenjo-Daro and Harappa, was later sustained by Mackay who, with specific reference to the round variant, added: 'It is extremely probable that slight variations distinguished objects of the same type from the different cities of the Indus Valley' (Mackay 1948: 343), thereby suggesting, as Gadd had done before, that the round form could very well have been a special trademark of some unexplored Harappan community.

In the years that followed, the amount of evidence pertaining to the question of the origin of the circular seals increased substantially as Oppenheim published his seminal article 'The seafaring merchants of Ur' (1954), and Bibby (1958) and Glob (1960: 212) not long after presented a number of circular seals from Bahrain and Failaka respectively. In commenting on the seals published by Bibby (1958), Wheeler sustained the opinion expressed by Gadd and Mackay but, in light of the new discoveries from

Bahrain and Oppenheim's convincing account on the alik Tilmun, he ventured to suggest a number of alternative sources for the circular seals: 'They came therefore, either overland through Persia or from the Persian Gulf — or, more probably, as their diversity suggest, were made at various entrepôts (such as Bahrain itself) of a cosmopolitan Persian Gulf trade of the kind which has been analysed by A.L. Oppenheim from Larsa tablets...' (Bibby 1958: 246 [Wheeler's comments]).

Regardless of the new finds from Bahrain, however, Hallo and Buchanan were reluctant fully to abandon the idea that the round seals with Indus inscriptions were made by 'Indians' but proposed the notion that the round form should be associated with the maritime trade (Hallo & Buchanan 1965: 205).

From this time onwards opinions vary but generally pivot around Bahrain as — if not *the* source — one of the most important nodes for traders employing the first circular seals. With reference to the circular seals with Indus text, Potts, for example, argued for an intimate relation between these seals and Bahrain: 'It suggests, for example, that a connection existed between Bahrain and the Indus Valley which may have contributed

to the very origin of sealing in the Arabian Gulf region' (Potts 1990: 165). Kjærum was cautious in his judgment when considering the provenance of his equivalent Arabian Gulf Type and circular seals with Indus text: 'Whether this seal group as a whole was developed on Bahrain and thence spread to Mesopotamia, Iran and the Indus region, like the proto-Dilmun and Dilmun seals, [...], is still an open question' (Kjærum 1994: 344). Parpola, on the basis of the non-Harappan sequences in the inscriptions on seals found in Bahrain and the fact that inscribed seals appear in burial contexts in Bahrain, suggested that the seals '...definitely point to an acculturation of Harappan traders.' (Parpola, 1994a: 315). Conversely, when Parpola focused on the characteristic circular shape alone, he suggested an Iranian source, noting that 'Perhaps it was in Iran (Susa) that the Near Eastern Indus traders first adopted the local circular form for their seals, which were then to become the models of the Gulf seals'. (1994a: 315). Recently other scholars (e.g. Vidale 2004; 2005) have proposed a host of new interpretations, in particular concerning the circular seals with Indus inscriptions, but those will be dealt with in more detail below.



Fig. 1.

Impressions and drawings of Gulf Type seals with Indus text and bull motif found in Early Dilmun burial mounds on Bahrain: a. Table 1. no. 10; b. Table 1. no. 11; c. Table 1 no. 56. (Scale: 150%).

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New seals with Indus text

Before addressing the typology of the seals in question the existing corpus of circular seals with Indus text are supplemented by two seals excavated by the Bahrain National Museum in 1999 and 2005 and a seal with Indus characters previously published by Al-Sindi (1999: no. 160) (Fig. 1/a–c). The latter was excavated by the same museum in 1983–84 and is illustrated here because of its particular relevance to the present study. The three seals are listed in Table 1 (where some basic information can be found) as nos. 10, 11 and 56 and were all recovered in Late Type mounds located in, respectively, the Janabiyah (a) and the Karzakkan (b and c) mound cemeteries (for a map of mound cemeteries see Laursen 2008: fig. 4).

All three seals are made of steatite with a white 'glaze', and they have the classic pierced boss with a single groove perpendicular to the perforation. The functional side of the stamp depicts the typical

short-horned bulls in profile standing with lowered head facing right on the impression. The upper half of the seals, which conventionally bears the short inscription in Indus characters when it is present, answers to this principle only to a varying degree in the three seals. Seal a (no. 10), which is rather worn, has a sequence of what appears to be at least six characters in the Indus script and it thus seems to be in full accordance with the standard scheme.² Seal b (no. 11) is more extraordinary because it is clearly inscribed with a string of pseudo-script, imitating that of the Indus Valley. In this seal a relatively clear 'twin' sign can be read as the first sign from the right, whereas the remaining area usually reserved for characters is filled with random strokes that follow the curvature of the seal, thus obviously intended to mimic text. The last seal c (no. 56) also deviates from the classic scheme, but is included here because two 'V' signs, reminiscent of the sign most commonly employed in native Indus inscriptions (see Parpola 1994b: fig. 5.1 no. 311), are used as fill in a composition that evokes the type with text.³ The seal shows two bulls placed back to back with the largest facing right on the impression, with the two 'V' signs inserted above each of their lowered heads. In this seal the smaller of the two bulls occupies the space otherwise reserved for text, providing us with a possible orientation of the seal.

Typological analysis

As stated in the introduction, this study focuses on the genesis and earliest phases of indigenous sealing technology in Dilmun culture. For this reason, evidence for developments of later 'generations' of 'endemic' seals such as the Proto-Dilmun and Dilmun Type proper (for a definition see Kjærø 1980; 1994) will be given only minor attention. The emphasis is on exploring the earliest circular stamp seals as a group, based on the idea that when analysed collectively it will communicate some of the complexity involved in the transmission of sealing technology into the Dilmun orbit.

The diverse group of circular seals which are believed to have served as inspiration for the later

² Asko Parpola has in a personal communication proposed the following tentative reading of seal no. 10. The sign numbers are those of his latest sign list (Parpola 1994b: 70–78, fig. 5.1). Starting from the right on the impression: (1) either very uncertain 53, or an unidentified sign (possibly badly drawn 'fish', 60) followed by 147, (2) 364, (3) 145, (4) either very uncertain 126 (see text 5 in Parpola 1994a: fig. 1718, where it follows 145 or very uncertain 125 ± very uncertain 128), (5) uncertain 16, (6) 145. Parpola tends to take (5) as 16 because this sign is rare, yet occurs quite often in the Near East (1994a: 306–307, texts nos. 6–8, 31 and 35) and the sequence 16–145 is in addition found at the beginning of the text on sealings K-69 to K-75 from Kalibangan (Joshi & Parpola A 1987: 312–313). At first sight, the right-hand 'man' sign of these 'twins' looks like the 'bowman' sign 38, but the lowest part of the would-be 'bow' seems to be the fourth short vertical stroke of the lowest row of the sign 145, and the remaining part of the 'bow' may be the roughly drawn sign 125 or 128. The sign 125 also occurs frequently on the Near Eastern Indus seals (see Parpola 1994a: nos. 29 [twice, once after 145], 18, 34, 39, 15 and 36). Seal 31 from Luristan contains two of the signs on the new seal (seal no. 10 in the present paper), namely 16 and 364. These two signs occur together in the unicorn seal with the three-sign inscription M-798 from Mohenjo-Daro (Shah & Parpola A 1991: 68). In the Near Eastern Indus texts, the sign 145 is also found in Parpola 1994a: nos. 5, 8 and 29. To summarise, there seems to be one sequence (16–145) also occurring in the Indus Valley (Kalibangan), while three of the signs occurring on this new seal, 364, 145 and 16, all seem to be somehow specifically related to the Near East or Near Eastern contacts/trade. The two 'man' signs together may be 'twins' or a 'couple'.

³ In contrast to the typical double projections usually found on the 'V' signs, those found on seal no. 56 only appear to feature single projections.

Table 1. Information on the 121 seals corresponding to the Gulf Type definition (Kjærum's Arabian Gulf Type). The seal numbers refer to the system used in this paper only. The seals found in a burial mound in Bahrain have been assigned a BBM number (Bahrain Burial Mound number in GIS), in those cases where the seal can be linked to a specific mound in the geographical information system of the Bahrain burial mound project. Reference to relevant literature is provided for each seal in the table.

Seal No.	Reference	Al-Sindi 1999 no.	Disc Height	Boss Height	Collar Width	Boss diameter * 2	Grooves	Gulf Type	Area/Site	Bahrain Burial Mound no. (BBM no)
1	Mackay 1943: pl. LI 23/ CISI vol.1	N/A	6.5	7.5	20	13		Gulf INDUS	Chanhu-Daro	
2	Marshall 1931: pl. CX.309	N/A	7.5	7.5	10.9	12	1	Gulf INDUS	Mohenjo-Daro	
3	Marshall 1931: pl. CXIV.478	N/A	7.18	9.82	18	14.5	1	Gulf INDUS	Mohenjo-Daro	
4	Marshall 1931: pl. CXII.383	N/A	9.45	8.55	18.5	14.5	1	Gulf INDUS	Mohenjo-Daro	
5	Mackay <i>et al.</i> 1937–38: pl. XCVL no 500	N/A	6	=D6-E6	15	14	1	Gulf INDUS	Mohenjo-Daro	
6	Kjærum 1994: fig.1726		8	6	13	14	1	Gulf INDUS	Qala'at al-Bahrain	
7	Kjærum 1994: fig.1725	279	10	5	15.5	13	1	Gulf INDUS	Qala'at al-Bahrain	
8	Srivastava 1991: fig. 55, left	180	10	5	12	16	1	Gulf INDUS	Karzakkan Cemetery	18839
9	Al-Sindi 1999: no. 182	182	8	6	13	14	1	Gulf INDUS	Saar Cemetery	N/A
10	This paper	N/A	10.6	5.4	11.6	18.9	1	Gulf INDUS	Janabiyah Cemetery	N/A
11	This paper	N/A	7	4.3	14.1	11.7	1	Gulf INDUS	Karzakkan Cemetery	14569
12	Kjærum 1983: cat. no 319	N/A	4.5	4.5	10	10	1	Gulf INDUS	Failaka	
13	Kjærum 1983: cat. no 279	N/A	10	5	13	20	3	Gulf INDUS	Failaka	
14	Amiet 1972: no. 1643	N/A	3	6	12	10	1	Gulf INDUS	Susa	
15	Amiet 1973: pl. 23a-b	N/A	0.5	11.4	14	11	1	Gulf INDUS	Luristan	
16	Gadd 1932: pl. I, no. 2	N/A	8.7	7.1	10.65	14.85	1	Gulf INDUS	Ur	
17	Gadd 1932: pl. I, no. 3	N/A	10.5	5.5	13.5	13.9	1	Gulf INDUS	Ur	
18	Gadd 1932: pl. I, no. 4	N/A	7.6	3.75	12.5	13	1	Gulf INDUS	Ur	
19	Gadd 1932: pl. I, no. 5	N/A	9	5.5	11	14	1	Gulf INDUS	Ur	
20	Gadd 1932: pl. I, no. 15	N/A	N/A	N/A	N/A	N/A	N/A	Gulf INDUS	Ur	
21	Gadd 1932: pl. I, no. 16	N/A	9.1	6.3	8.55	14.7	1	Gulf INDUS	Ur	
22	Sarzac & Heuzey 1884-1912: pl. 30.3a-b	N/A	5	5	14	10	2	Gulf INDUS	Girsu	
23	Gadd 1932: pl. I, no. 17	N/A	7.5	3.5	10.95	14.4	N/A	Gulf INDUS	Near East	
24	Gadd 1932: pl. I, no. 18	N/A	13.9	6.3	13.1	20.1	1	Gulf INDUS	Near East	
25	Langdon 1932: p. 48 (J. Rosen collection)	N/A	6.5	5.5	13	11.5	1	Gulf INDUS	Mesopotamia?	
26	Buchanan 1981: no. 1088/Newell 23	N/A	9	4.5	12.5	11	1	Gulf INDUS	Mesopotamia?	
27	Buchanan 1981: no. 1089/Newell 876	N/A	8.6	3.4	8.4	13.8	1	Gulf INDUS	Mesopotamia?	
28	Winkelmann 1999: abb. 2	N/A	7	7	15	10	0	Linear-Elamite	western Iranian plateau	
29	Kjærum 1983 cat. no. 294	N/A	7	4.5	9	15	2	Gulf Type	Failaka	
30	Kjærum 1983: cat. no. 295	N/A	7.5	3.5	8	15	3	Gulf Type	Failaka	
31	Al-Sindi 1999: no. 284	284	N/A	N/A	N/A	N/A	N/A	Gulf Type	Karzakkan Cemetery	20393
32	Al-Sindi 1999: no. 163	163	11	4	6.5	13.5	0	Gulf Type	Karzakkan Cemetery	20038
33	Al-Sindi 1999: no. 162	162	N/A	N/A	N/A	N/A	N/A	Gulf Type	Karzakkan Cemetery	75306
34	Al-Sindi 1999: no. 137	137	13.2	3.8	9.8	15	3	Gulf Type	Rifa mounds	N/A
35	Al-Sindi 1999: no. 98	98	N/A	N/A	N/A	N/A	N/A	Gulf Type	Karzakkan Cemetery	N/A
36	Al-Sindi 1999: no. 89	89	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Cemetery	N/A
37	Al-Sindi 1999: no. 8	8	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Cemetery	N/A

Table 1. (Continued)

Seal No.	Reference	Al-Sindi 1999 no.	Disc Height	Boss Height	Collar Width * 2	Boss diameter	Grooves	Gulf Type	Area/Site	Bahrain Burial Mound no. (BBM no)]
38	Al-Sindi 1999: no. 130	129	N/A	N/A	N/A	N/A	N/A	Gulf Type	Hamad Town	N/A
39	Al-Sindi 1999: no. 133	133	N/A	N/A	N/A	N/A	N/A	Gulf Type	Hamad Town	N/A
40	Al-Sindi 1999: no. 166	166	N/A	N/A	N/A	N/A	N/A	Gulf Type	Al Maqsha	N/A
41	Al-Sindi 1999: no. 134	134	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Cemetery	N/A
42	Al-Sindi 1999: no. 129	129	N/A	N/A	N/A	N/A	N/A	Gulf Type	Hamad Town	N/A
43	Al-Sindi 1999: no. 181	181	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Cemetery	N/A
44	Al-Sindi 1999: no. 179	179	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Cemetery	N/A
45	Al-Sindi 1999: no. 159	159	N/A	N/A	N/A	N/A	N/A	Gulf Type	Hamad Town	N/A
46	Mughal 1983: pl. XIV no. 5	125	N/A	N/A	N/A	N/A	N/A	Gulf Type	Sar Burial complex	
47	Mughal 1983: pl. XIV no. 4	127	N/A	N/A	N/A	N/A	N/A	Gulf Type	Sar Burial complex	
48	During Caspers, 1977	N/A	6	3	7	14	1	Gulf Type	Stray find Diraz	
49	This paper	N/A	9	4	10	13	1	Gulf Type	Karzakkan Cemetery	20160
50	Ibrahim 1982: pl. 61.2	14	7	5	8	14	1	Gulf Type	Sar Burial complex	
51	Al-Sindi 1999: no. 174	174	9.5	2.5	9	13.5	2	Gulf Type	Karzakkan Cemetery	14337
52	McNicoll and Roaf: 1975 pl. III:B-C	128	5	5	8	12	1	Gulf Type	Diraz East	
53	Al-Sindi 1999: no. 138	138	9	2.5	6	10	1	Gulf Type	Karzakkan Cemetery	19081
54	Al-Sindi 1999: no. 298	298	N/A	N/A	N/A	N/A	N/A	Gulf Type	Janabiyah Cemetery	N/A
55	Al-Sindi 1999: no. 164	164	N/A	N/A	N/A	N/A	N/A	Gulf Type	Aali Cemetery	N/A
56	Al-Sindi 1999: no. 160	160	10	4.5	11	14	3	Gulf INDUS	Karzakkan Cemetery	20362
57	Al-Sindi 1999: no. 65	65	9.5	5.5	14	13.5	1	Gulf Type	Karzakkan Cemetery	N/A
58	Al-Sindi 1999: no. 71	71	9	3	10	15	1	Gulf Type	Saar Cemetery	N/A
59	Al-Sindi 1999: no. 126	126	N/A	N/A	N/A	N/A		Gulf Type	Saar Cemetery	N/A
60	Al-Sindi 1999: no. 165	165	0	11	21	0	0	Gulf Type	Saar Cemetery	N/A
61	Al-Sindi 1999: no. 86	86	10.5	N/A	N/A	13øsen?	N/A	Gulf Type	Karzakkan Cemetery	N/A
62	Srivastava 1991: fig. 55, right	10	8.5	5.5	8	18	1	Gulf Type	Karzakkan Cemetery	18789
63	Al-Sindi 1999: no. 136	136	7	4	7	13.5	1	Gulf Type	Sar Burial complex	
64	Ibrahim 1982: pl. 61.3	176	8.5	4	7	13	0	Gulf Type	Sar Burial complex	
65	Ibrahim 1982: pl. 60.4	N/A	7.7	4.6	10	13.8	1	Gulf Type	Saar Cemetery	N/A
66	Ibrahim 1982: pl. 60.3	295	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Cemetery	N/A
67	Mughal 1983: pl. XIV no. 2	296	N/A	N/A	N/A	N/A	N/A	Gulf Type	Sar Burial complex	
68	Kjærum 1994: fig.1731	140	11	5	13.5	14.5	1	Gulf Type	Qala'at al-Bahrain	
69	Kjærum 1994: fig.1727	283	12	5	10	15	0	Gulf Type	Qala'at al-Bahrain	
70	Kjærum 1994: fig.1732	177	7.5	3.5	8	13	1	Gulf Type	Qala'at al-Bahrain	
71	Kjærum 1994: fig.1728	297	8	N/A	13	9	N/A	Gulf Type	Qala'at al-Bahrain	
72	Kjærum 1994: fig.1729	172	10.5	4	6.5	13.5	2	Gulf Type	Qala'at al-Bahrain	
73	Kjærum 1994: fig.1734	171	N/A	N/A	N/A	N/A	N/A	Gulf Type	Qala'at al-Bahrain	
74	Al-Sindi 1999: no. 2	2	11	5	11	16.5	2	Gulf Type	Saar Cemetery	N/A

Table 1. (Continued)

Seal No.	Reference	Al-Sindi 1999 no.	Disc Height	Boss Height	Collar Width * 2	Boss diameter	Grooves	Gulf Type	Area/Site	Bahrain Burial Mound no. (BBM no)]
75	Kjærum 1994: fig.1733	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	Qala'at al-Bahrain	
76	Kjærum 1994: fig.1730	N/A	7.5	2	8.5	12	2	Gulf Type	Qala'at al-Bahrain	
77	Al-Sindi 1999: no. 310	310	8	4	7	15	2	Gulf Type	Karanah	
78	This paper	N/A	N/A	N/A	N/A	N/A	0	Gulf Type	Karzakkan Cemetery	18207
79	Al-Sindi 1999: no. 237	237	10.5	5.5	8	17	3	Gulf Type	Karzakkan Cemetery	18207
80	unpublished		9	4	7	16	1	Gulf Type	Janabiyah Cemetery	N/A
81	unpublished		5	5	5	13	0	Gulf Type	Janabiyah Cemetery	N/A
82	Kjærum 2007: no. 1	N/A	N/A	N/A	12	17	0	Gulf Type	"Charnel House"	
83	Kjærum 2007: no. 2	N/A	N/A	N/A	7	15	0	Gulf Type	"Charnel House"	
84	Kjærum 2007: no. 3	N/A	N/A	N/A	11	15	3	Gulf Type	"Charnel House"	
85	Kjærum 2007: no. 4	N/A	N/A	N/A	10	13	1	Gulf Type	"Charnel House"	
86	Kjærum 2007: no. 5	N/A	N/A	N/A	7	16	1	Gulf Type	"Charnel House"	
87	Kjærum 2007: no. 6	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	"Charnel House"	
88	Kjærum 2007: no. 7	N/A	N/A	N/A	11	14	2	Gulf Type	"Charnel House"	
89	Kjærum 2007: no. 8	N/A	N/A	N/A	10	11	0	Gulf Type	"Charnel House"	
90	Kjærum 2007: no. 9	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	"Charnel House"	
91	Kjærum 2007: no. 10	N/A	N/A	N/A	N/A	N/A	1	Gulf Type	"Charnel House"	
92	Kjærum 2007: no. 11	N/A	N/A	N/A	5	11	0	Gulf Type	"Charnel House"	
93	Kjærum 2007: no. 12	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	"Charnel House"	
94	Kjærum 2007: no. 13	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	"Charnel House"	
95	Kjærum 2007: no. 14	N/A	N/A	N/A	5	14	0	Gulf Type	"Charnel House"	
96	Kjærum 2007: no. 15	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	"Charnel House"	
97	Crawford 2001: no. 2622:05	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Settlement	
98	Crawford 2001: no. 4197:03	N/A	N/A	N/A	N/A	N/A	1	Gulf Type	Saar Settlement	
99	Crawford 2001: no. 4139:01	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Settlement	
100	Crawford 2001: no. 4300:01	N/A	N/A	N/A	N/A	N/A	3	Gulf Type	Saar Settlement	
101	Crawford 2001: no. 5506:05	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Settlement	
102	Crawford 2001: no. 6581:02	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Settlement	
103	Crawford 2001: no. L18:27:07	N/A	N/A	N/A	N/A	N/A	N/A	Gulf Type	Saar Settlement	
104	This paper	N/A	9.9	2.1	8.7	13.2	1	Gulf Type	Karzakkan Cemetery	75548
105	This paper	N/A	7.7	N/A	10.2	13	1	Gulf Type	Karzakkan Cemetery	23522
106	This paper	N/A	9.4	3.4	4.8	14.6	1	Gulf Type	Karzakkan Cemetery	23047
107	This paper	N/A	9.5	3.1	10	14.2	1	Gulf Type	Karzakkan Cemetery	22363
108	This paper	N/A	8.1	7	24.8	1.7	1	Gulf Type	Karzakkan Cemetery	23559
109	This paper	N/A	8.8	3.5	7.3	13.7	1	Gulf Type	Karzakkan Cemetery	15309
110	This paper	N/A	6.2	3.5	4.1	12.9	3	Gulf Type	Karzakkan Cemetery	14784

Table 1. (Continued)

Seal No.	Reference	Al-Sindi 1999 no.	Disc Height	Boss Height	Collar Width * 2	Boss diameter	Grooves	Gulf Type	Area/Site	Bahrain Burial Mound no. (BBM no.)
111	This paper	N/A	8.4	3.5	11.4	12	1	Gulf Type	Karzakkan Cemetery	75638
112	This paper	N/A	10.5	4.3	7.6	12.8	1	Gulf Type	Karzakkan Cemetery	15473
113	This paper	N/A	10.4	4.3	6.9	14.7	1	Gulf Type	Karzakkan Cemetery	75621
114	This paper	N/A	10.3	3.7	10	12.8	1	Gulf Type	Karzakkan Cemetery	22010
115	This paper	N/A	10.5	5.2	8	15.5	1	Gulf Type	Karzakkan Cemetery	20160
116	This paper	N/A	9.8	5.2	10.2	13.4	1	Gulf Type	Karzakkan Cemetery	15557
117	Beyer 1989: no. 249	N/A	7.5	5.5	10	14	2	Gulf Type	Bahrain unsp.	
118	Presinger 1983: fig. 186/11	N/A	4	5	5.5	14.5	0	Gulf Type	Dahran South	
119	Lamberg-Karlovsky 1970 fig. 4.6	N/A	8.2	5.8	10	15	0	Gulf Type	Tepe Yahya	
120	Barger 1969: 139-140	N/A	N/A	N/A	N/A	N/A	1	Gulf Type	Al Khobar	
121	Zarins 1978: pl. 70/583	N/A	6	4.5	7	14	1	Gulf Type	Tarut Island	

development of the more homogeneous Dilmun type have thus far been referred to by different names, the most common being Persian Gulf seals (Bibby 1958 [Wheeler's comments]) and the Arabian Gulf Type (Kjærø 1980; 1983; 1994). Depending on the context in which they occur these two types often cover different seal definitions, but they are still mistakenly treated as one and the same. To get rid of this confusion a broad and simple definition of a seal termed 'Gulf Type' is proposed and formulated here to encompass the above-mentioned circular stamp seals of pre-Dilmun types (this Gulf Type fully corresponds with Kjærø's Arabian Gulf Type but not always to the definition used for the Persian Gulf Type). The 'Gulf Type' should not be confused with either Hallo and Buchanan's 'Gulf Seal' (1965: 204–209) or Mitchell's use of the term 'Gulf' in his grouping of the seals (1986: 278).

The Gulf Type is here defined by a circular disc, which on one side holds the functional stamp motif while centred on the other side is a raised semi-spherical knob. The circular (occasionally oval) knob, which is usually referred to as the 'boss', is pierced horizontally for suspension, and perpendicular to this perforation it is typically decorated with one, or much more rarely, two or three grooves. The type is predominantly made of steatite covered with a white 'glaze' although specimens in other stones, shell and ivory have occasionally been reported.

For the present study 121 seals corresponding to these criteria have been recorded, and although seals found more recently are currently stored in the Bahrain National Museum, the sample has been regarded as sufficiently representative to enable qualified statements on the artefact category as a whole. Here, analysis of the Gulf Type seals is equally aimed at identifying formal and qualitative structures in the material from the expectation that both dimensions will help break the type down into a number of culturally significant variants.

In order to introduce some structure to the presentation of the typological analysis it has been organised into a number of sub-sections, the first of which investigates morphological variation. Thereafter a second section looks into the glyptic, iconographic and stylistic variation and a third addresses the matter of the epigraphy found on a number of the Gulf Type seals. The typological analysis is concluded with a section in which the question of

chronology is addressed, after which there follows a discussion that begins with an examination of the patterns in the geographical distribution of the seals.

The seals

Information on all 121 seals included in this examination is listed in Table 1 where references to relevant literature can be found together with other elementary data. In addition to the three 'new' seals with Indus characters presented above, the present writer has had the opportunity to examine all seals in the Bahrain National Museum found in the Karzakkan and Buri mound cemeteries, which in the literature collectively have been identified as the Hamad Town excavations. In this material a total of seventeen seals, conforming to the Gulf Type definition and which have not previously been published, were recorded. In Table 1 these unpublished seals, which all derive from burial mounds in the Karzakkan mound cemetery, have been assigned the following numbers: 49, 78, 80–81 and 104–116. Since they have not previously been made available to the public, photographs of impressions and seal reverse are shown at the end of the article for the twelve seals that are not too fragmented (see Fig. 17).

One seal (Table 1, no. 13) from Failaka Island, Kuwait (Kjærøum 1983: no. 279), which, with three grooves and four dots-in-circles on the boss, is technically of the Dilmun Type, is included in the sample because it bears what appears to be a (pseudo-?)inscription in Indus characters. Another atypical seal allegedly from the western Iranian plateau (Table 1, no. 28) which was previously published by Winkelmann (1999: abb. 2) is also included in the dataset because the seal exhibits the classic composition which Potts has labelled 'bull in profile with Indus text' (1990: 165), but where the conventional Indus characters are substituted by an inscription in Linear-Elamite.⁴

Morphological variation

Probably as a consequence of the relatively small number of seals, little attention has previously been paid to the morphological variation which characterizes the Gulf Type seals. However, with the larger material presently available it is of relevance to explore whether morphological groupings in this relatively homogeneous corpus can be identified and if so, whether these can perhaps be related to other variables such as the presence/absence of text, geographical provenance and variation in glyptic style and iconography. It should be noted that the earliest type of Gulf seals were undoubtedly used by merchants for administrative purposes, although sealings and other impressions associated with these early seals have yet to be discovered.

The underlying assumption behind the present approach is basically that if the spatial distribution of any such identified variants of the Gulf Type differ, it can safely be expected that either the orbit in which it circulated and/or its place of production almost certainly also differed. Further, if any variant exhibits a distinct geographical cluster this can, depending on the general circumstances, be regarded as being suggestive of, or identical with, a shared place of production and circulation. Finally, if seals belonging to one Gulf Type variant are found clustered in more than one geographical area it is expected that their owners must also predominantly have operated and/or resided in those locales. Geographical patterning is thus regarded as one of only a few strong arguments that can confirm the cultural significance of a particular variant identified. Aspects which indicate that matters of chronological difference are contributing to an observed variation are obviously also to be taken into account in any interpretation.

Principal components analysis (PCA) has been chosen as the method with which to identify structures in the morphological variation of the seals because it is multivariate and aimed precisely at identifying patterns of covariance in a set of measured variables (Madsen 1988; 2007a: 1). PCA belongs to the group of so-called factor analyses of which correspondence analysis is the more commonly known, and is not a formal test as such but an exploratory tool which identifies 'hidden' metric

⁴ J.J. Glassner quite recently referred to an unpublished seal with Linear-Elamite inscription as being from Bahrain (Glassner 2008: 173, citing Eidem 1994: 303) but no such seal is mentioned by Eidem or has for that matter been found in Bahrain. The seal published by Glock (1988: no. 16) from the Bailey Collection is however identical with that published in 1999 by Winkelmann, which at that time was in the Ligabue collection in Venice. My thanks to Dr P. Lombard, CNRS Lyon, for his help with the literature.

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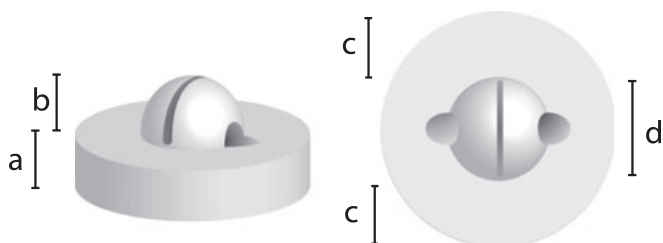


Fig. 2.

Idealised representation of a Gulf Type seal seen from the side (left) and vertically (right). The brackets indicate the measured variables: Disc Height (a), Boss Height (b), Collar Width (c) and Boss Diameter (d).

structures and thereby often provides an improved foundation for further interpretation.

The geometric shape of the Gulf Type seal is a simple one and in order to explore the morphological variation, the dimensions of Disc Height (a), Boss Height (b), Collar Width (c) and Boss Diameter (d) have been defined and recorded as morphological variables (Fig. 2). Even if four variables come close to the minimum of three required for a multivariate statistic, this says nothing about how suitable PCA is to the dataset in question. The dimensions have occasionally been reconstructed in those cases where seals were fragmented, based on the preserved parts, while in rare cases totally missing 'dimensions' have been estimated by subjective judgment (these measurements are marked in bold numerals in Table 1).

It has been possible to obtain the dimensions of seventy out of the 121 seals conforming to the Gulf Type definition.⁵ The software CAPCA, which is a free add-on to Microsoft excel (Madsen 2007b) has been used for the analysis because it allows for the performance of principal component analysis (as well as correspondence analysis and metric scaling) in a software environment that is familiar to most. Only the four metric variables presented above were used for the analysis. The setting chosen to perform the PCA was the correlations matrix (Persons *r*) (*contra* covariance matrix). The function *Automatic transformation* was activated to obtain normality in the sample in order to reduce the direct influence of size variation because it could be supposed, for

example, that the presence or absence of an inscription may be related to the overall size of the seal stamp. The diameters of the seals with inscriptions (26.66 ± 1.44 , 95% CI, $n = 28$) and the seals without inscriptions (22.49 ± 0.77 , 95% CI, $n = 54$) are in fact significantly different ($W = 1235$, $p = 2.795e-06$, Wilcoxon Rank Sum Test)⁶. One could speculate that the difference in size simply reflects the need to make room for an inscription, but as we shall see below the structure observed in the PCA is not solely a product of different sizes but also the result of relative proportional variation in both disc height, boss diameter and boss height, which have no influence on the size of the functional side of the stamp.

As a result of the analysis, 79% of the variation in the four measured variables is explained by the distribution of the seals (objects) on the 'x' and 'y' axis alone (Fig. 3). The seals are distributed in an oblong scatter centred on 0.0 from which only a smaller number fans out in different directions.

At first impression this pattern appears to be of limited explanatory value but when attention is turned towards the presence/absence of an inscription a much clearer structure emerges (Fig. 3). With a few exceptions, it can be observed that seals bearing inscriptions are found in the negative end of the 'x' axis (86%), while the vast majority of uninscribed seals are located in the positive end (84%). This structure is extremely important because for the first time it allows a division of the Gulf Type seals into two major morphological groups, one of which was normally associated with Indus inscriptions.

The Gulf Type seals come from a vast geographical area encompassing Bahrain, the Indus Valley (Mohenjo-Daro and Chanhu-Daro), Iran (Kerman, Luristan, Susa and the western Iranian plateau), Kuwait (Failaka), Mesopotamia (Ur, Girsu, Babylon and others unspecified) and the U.A.E. (Tell Abraq). In a later section we shall look more closely at patterns in the geographical distribution of the seals, but this important dimension of variation will also be consulted here to gain a first hand impression of

⁵ The ivory seal (Gulf Type imitation) from Tell Abraq (Potts 2000: 122) which is technically of the Gulf Type was excluded for consistency.

⁶ Wilcoxon Rank Sum Text was used instead of Students *t*-test, since a Shapiro-Wilk normality test demonstrated that the diameters of seals with inscriptions departed slightly from normality ($W = 0.93$, p -value = 0.049).

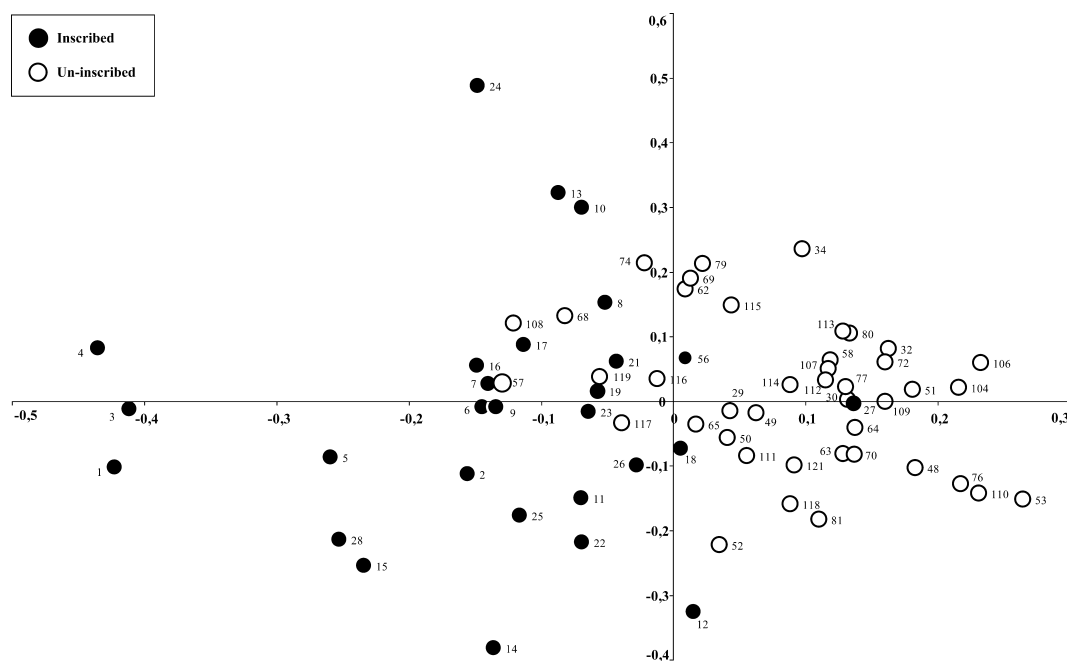


Fig. 3.

The distribution of objects (seals) on the 'x' and 'y' axes in the principal components analysis. White dots indicate seals without text, black dots seals with text. Object numbers refer to seal numbers listed in Table 1 (N = 70).

the overall cultural significance of the structures identified in the PCA.

When we look separately at the geographical provenance of the seals with inscriptions, most of which were shown to belong to a common morphological group, clear structures appear once more (Fig. 4). First of all, located furthest towards the negative end of the 'x' axis are four out of five of the seals from the Indus Valley (Group 1). This isolated position suggests that these seals belong to a morphological group so far diagnostic to this region. When attention is turned towards the three seals with inscriptions found in Iran, a small but relatively discrete group is encountered once more, this time in relative isolation centred on $-0.2/-0.3$ (Group 2). Although isolated, it should be noted that the group of 'Iranian' seals appears to bear some morphological resemblance to the Indus Valley group. Judging from the positions of the inscribed seals found in Bahrain and Mesopotamia, it appears that these seals on an overall level belong to one common morphological group, which clusters around $-0.1/0.0$ (Group 3). With a few exceptions this group is located at the negative end of the 'x' axis. Among these seals attention should

also be paid to the fact that, of the seven 'Near Eastern' seals with provenance confirmed to Mesopotamian cities — including Ur (nos. 16–19, 21), Girsu (no. 22) and Babylon (no. 23) — all are in the PCA found in close proximity to the Bahraini specimens within the Group 3 cluster. Additionally, it is noteworthy that seal no. 2 from Mohenjo-Daro adheres more closely to the Group 3 seals and not as would be expected to the other Indus Valley seals found in Group 1. It is furthermore significant that the two seals from Failaka (Fig. 4/12–13) clearly represent outliers together with a few of Mesopotamian provenance, including no. 24, an extraordinarily large and thick seal. Finally, it is apparent that inscribed seals from Bahrain and Mesopotamia (Group 3) are morphologically different from, but associated with, the bulk of un-inscribed seals found clustered at the positive end of the 'x' axis (Group 4). How the association between the two related groups should be interpreted poses a challenge, but it is striking that with only six exceptions — including seals from Tepe Yayha (Table 1: no. 119), Failaka (Table 1: nos. 29–30), Dhahran and Tarut (Table 1: nos. 118 and 121) and the ivory specimen from Tell Abraq (Potts

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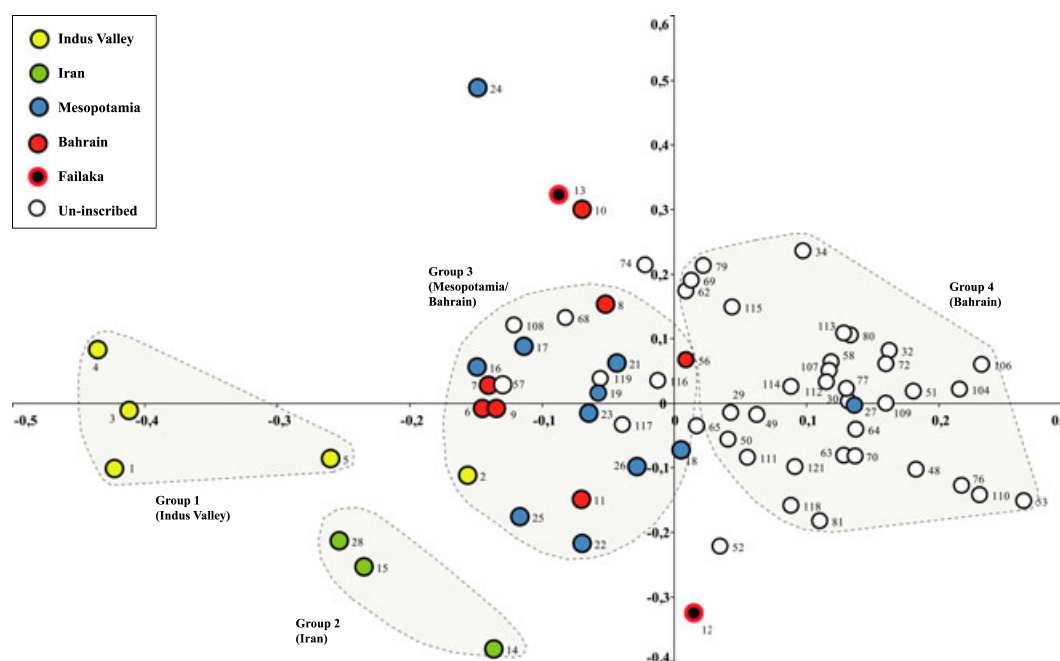


Fig. 4.

The distribution of objects (seals) on the 'x' and 'y' axes in the principal components analysis. The inscribed seals have been coloured according to overall geographical provenance, the uninscribed seals, the majority of which come from Bahrain, have been left uncoloured. Four morphological groups are suggested on the basis of a combination of the position in the principal components analysis and corresponding patterns in geographical provenance. Object numbers refer to seal numbers listed in Table 1 (N = 70).

2000: 122) — the uninscribed Gulf Type seals all originate from Bahrain Island.⁷

To sum up, the majority of the Gulf Type seals fall within four morphological groupings identified in the PCA, and it was observed that the seals with inscriptions belong to three groups each of which can be associated with, respectively, the Indus Valley, Iran and Bahrain/Mesopotamia.

Until now, the actual dimensions which characterise the seals that fall into each of the four groups, including those without inscriptions (relevant for Groups 3 and 4 only), have been disregarded, which is why the focus is now turned towards this aspect. The position on the 'x' and 'y' axis of the four morphological variables (Fig. 5) reveals a strong covariance between Boss Height and Collar Width on the one hand and Boss Diameter and Disc Height on the other. The fact that the variables covariate in this manner shows that the variation is not a product of the larger average size of the inscribed seal, and

Variables on 1. and 2. principal axes

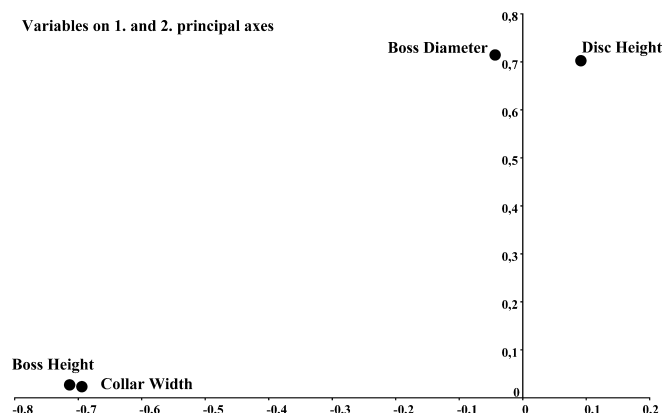


Fig. 5.

The distribution of the four measured variables: Boss Height, Collar Width, Boss Diameter and Disc Height on the 'x' and 'y' axes in the principal components analysis.

consequently it can be ruled out that structures observed in the PCA merely reflect the difference in diameter between inscribed and uninscribed seals, in which case one would expect Collar Width and Boss Diameter to covariate.

In general, quite subtle and gradual differences characterise the morphological variation of the seals, and the small distinctions that contribute to the

⁷ To this number one can probably safely add the fifty Gulf Type seals from Bahrain that were excluded in the PCA due to insufficient metric information.

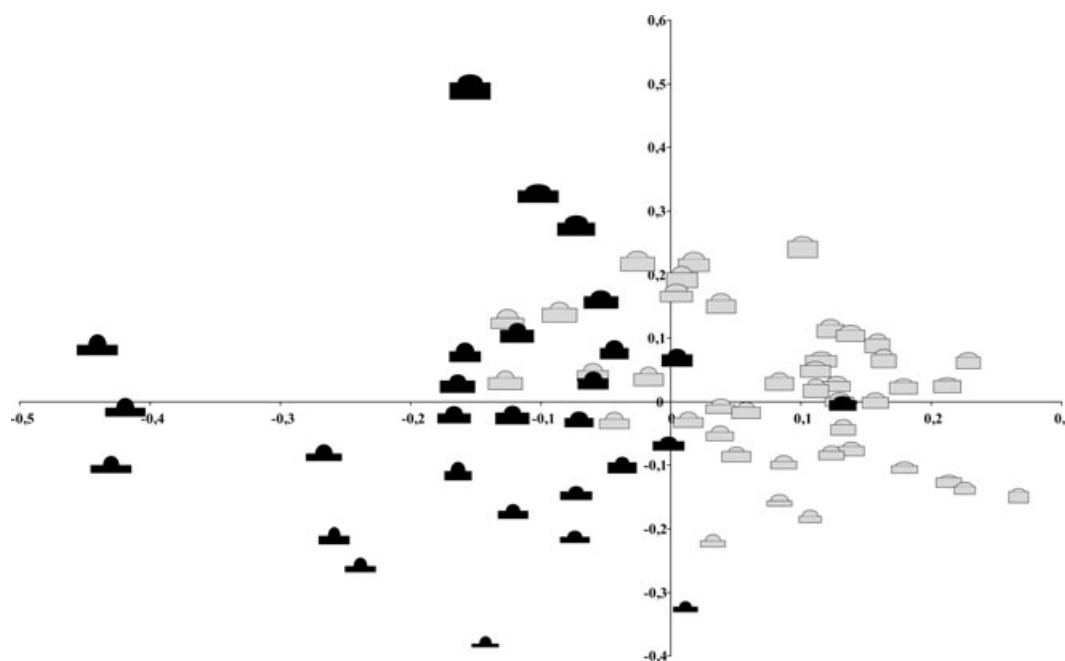


Fig. 6.

The distribution of objects (seals) on the 'x' and 'y' axes in the principal components analysis. The contour of each seal is illustrated as a profile section in relative scale. Inscribed seals are shown in black, uninscribed seals in grey. Object numbers refer to the seal numbers listed in Table 1 (N = 70).

structures observed in the PCA are not easily communicated in print. A drawing of each seal (object) is illustrated in the PCA plot as an idealised section in relative scale (Fig. 6) in order to provide an impression of the relative differences that characterise seals of Gulf Type.

For the purpose of creating a provisional typology of the seals that takes into account not only the shape but also the obvious significance of both the presence/absence of script and the geographical provenance, the four groupings identified above are used here to summarise the morphological information into four 'type-distinct' categories of seals created by the metric average of those seals which belong to each of the four groups (Fig. 7). Group 1 is comprised of large seals characterised by a relatively thin disc with a large diameter and a wide collar (Fig. 7/a). The boss is relatively high and has a small diameter. Group 2 shares a number of similarities with Group 1 but its seals are significantly smaller and differentiated by a thinner disc and a slightly lower boss (Fig. 7/b). Seals of Group 3 are relatively large but have a boss that is relatively low and significantly wider than those of the Group 1 and 2 seals (Fig. 7/c). Group 3 seals are also distinguished

from seals of Group 1 and 2 by generally having a much thicker disc. Group 4 seals have the lowest and widest boss of all and the disc is thicker than on seals of all other groups (Fig. 7/d). It should be noted that the wide boss of Group 4 seals is the one that most closely corresponds to that which can be found on the later Dilmun Type seals. This also shows that within the morphological variation of the Gulf Type one can apparently observe continuity with the Dilmun Type (Kjærø 1983; 1994: fig. 1723; Højlund 2000: fig. 1).

Glyptic, iconographic and stylistic variation

In this section, which looks at glyptic, iconographic and stylistic variation, the fifty-one seals, which were excluded from the PCA because of unavailable metric information, are whenever possible incorporated in the analysis. In some cases these seals, which have not been formally associated with a particular morphological group through the PCA because they could not be measured, will however be dealt with under one of the four morphological groups with which, for the sake of argument, they share the greatest iconographic resemblance.

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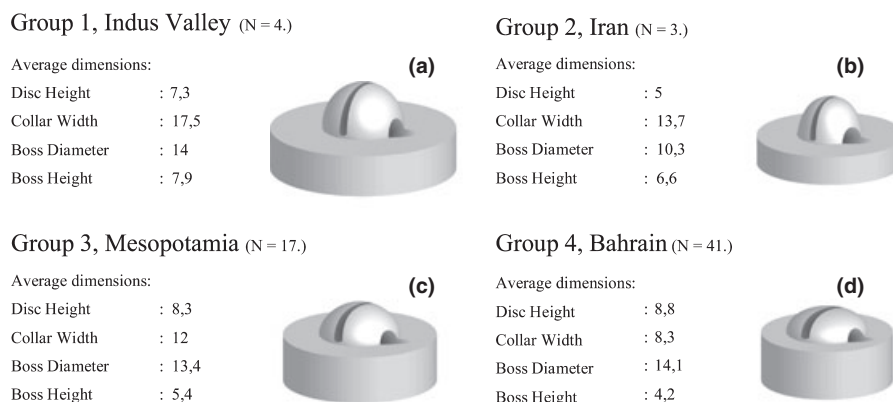


Fig. 7.

The average dimensions of the seals associated with each of the four morphological groups are given for the four measured variables Boss Height, Collar Diameter, Boss Diameter and Disc Height. A drawing idealising an 'average' seal is given for each of the four groups.

The lower Indus seals from Mohenjo-Daro and Chanhu-Daro that constitute Group 1 are rather heterogeneous in terms of glyptic motifs but stand to one side as a consequence of some features unique to this group, in general contrast to the glyptic motifs found on the rest of the Gulf Type sample (Fig. 8). These seals have occasionally been taken as direct evidence of western Gulf seals in the Indus Valley (see e.g. Reade 2008: 17), but a quick examination reveals that this is an incorrect assumption. On the contrary, this group finds its closest stylistic parallels among the classic square seals of the Harappan culture and it is different elements from this tradition which are not found on any other seals answering to the Gulf Type definition, that make the group stylistically distinct.

All four seals appear to be decorated with various versions of the 'bull' motif that is so frequently

found below the inscription on the classic square Indus Valley seals. In seals 1 and 5, the carving of the bulls accords perfectly with some of the most frequently recorded beasts in the Indus iconography, the unicorn and the short-horned bull. Judging from the preserved fragment of seal no. 3 (Fig. 8/3) (for an excellent photograph see Shah & Parpola A 1991: 179, M-1369 A) this seal also appears to have carried the 'unicorn' motif, of which a hatched upper neck and an ear can be distinguished. However, a bull (gaur) has also been suggested as a possibility by Vidale (2005: 150).

Seal no. 4 (Fig. 8/4) depicts a fantastic animal with six protomen-like heads concentrically radiating from a so-called 'whorl' composition or 'Wirbel' (Franke-Vogt 1991: 99; During Caspers 1994: 99), of which at least one is a 'unicorn' in the classic style with hatched neck and collar (see Rissman 1989 for

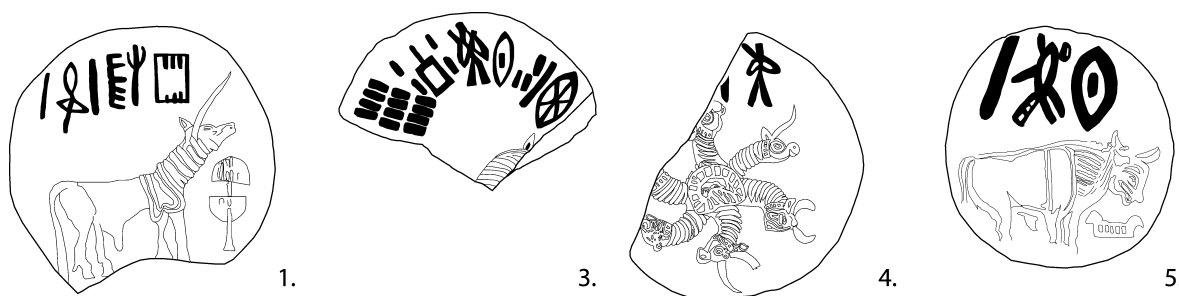


Fig. 8.

The lower Indus seals from Mohenjo-Daro and Chanhu-Daro that constitute Group 1, illustrated as impression drawings. The numbers refer to the seal numbers in Table 1 (Scale 1:1). (1. after Mackay 1943: pl. LI/23; 3. and 4. after Marshall 1931: pl. CXIV/478 and pl. CXII/383, respectively; 5. after Mackay *et al.* 1937–38: pl. XCVL/500).

Fig. 9.

The inscribed seals of Groups 2 and 3 illustrated as impression drawings. The numbers refer to seal numbers in Table 1. Outliers (nos. 24, 13, 10, 12) are included together with seal no. 20 that due to unavailable metric information was omitted in the PCA. (Scale 1:1). (2. after Marshall 1931: pl. CX/309; 16–21, 23 and 24. after Gadd 1932: pl. I/2–5 and pl. III/15–18.; 22. after Sarzec & Heuzey 1884–1912: 321–322 and pl. 30.3a–b; 25. Langdon 1932: p. 48; 26. after Hallo & Buchanan 1981: no. 1088; 27. after Buchanan 1981: no. 1089; 7. after Kjærøum 1994: fig. 1725; 8. after Srivastava 1991: fig. 55; 11. this article; 9. after Al-Sindi 1999: no. 182; 56. after Al-Sindi 1999: no. 160.; 6. after Kjærøum 1994: fig. 1726; 13. after Kjærøum 1983: no. 279 and drawing by Nancy Zeffert (technically Dilmun Type); 10. this article; 12. after Kjærøum 1983: no. 319 and drawing by Nancy Zeffert; 28. after Winkelmann 1999: Abb. 2.; 15. after Amiet 1973: pl. 23a–b; 14. Amiet 1972: pl. 153/1643).

the regional distribution of different ‘unicorns’). Other animals in the composition can be identified as various bulls and a tiger. The rendering of the ‘unicorn’ and the other protomen contribute to an impression of the glyptic style as native to the Indus Valley tradition. However, from an iconographic perspective the ‘whorl composition’ is a ‘foreign’ and rarely used element in the Indus tradition that has been argued to mirror the adaptation of a western prototype into the Indus iconography (Collon 1994: 222, fig. 32; During Caspers 1994: 99).⁸ From c.2000 BC onwards, the ‘whorl’ motif appears frequently on Dilmun Type seals on Bahrain and Failaka and also as far north-west as Acmhöyük in Anatolia, where it is found on an impression dated to the reign of Shamshi-Adad I of Assyria (c.1800 BC) (Collon 1994: 222; Porada 1971: pl.10/8).

If the ‘whorl’ motif were indeed an influence of Near Eastern affinity, this would fit admirably with its appearance on a seal of circular shape that is also alien to Mohenjo-Daro and the Indus Valley. Having hinted at the possible western influence, one should recall that all four seals from Group 1 depict creatures and elements well attested in the classic iconographic repertoire and which additionally are carved in glyptic styles characteristic of the Indus Valley tradition. This should in turn contrasted with the fact that ‘unicorns’ and ‘short-horned bulls’ rendered like these classic Indus types do not appear at all among the other Gulf Type seals. The seals in Group 1 are also distinguished from the other Gulf Type seals by having a

horizontal inscription such as those found on seals 1, 4 and 5 (Fig. 8/1, 4, 5). The horizontal arrangement of the Indus characters conforms to the tradition of the classic square Indus seals where text is always written in alignment with the straight upper edge of the square seal. The Indus inscriptions, which are found on Gulf Type seals of Groups 2 and 3, almost by convention follow the upper curve of the stamp.⁹

The ‘cult-stand/standard’ is yet another iconographic feature of Indus Valley affinity that is found exclusively on Group 1 seals; it appears on seal no. 1 (Fig. 8/1) in front of the ‘unicorn’ in the classic variant with rounded upper ‘cage’ and lower ‘bowl’. An educated guess would be that a similar object was present on the parts that have broken off from seal no. 3 (Fig. 8/3). The ‘manger’ of the kind seen below the bull’s head on seal no. 5 (Fig. 8/5) and which, judging from its repeated position on the Indus seals, must have shared a symbolism or at least connotations similar to those of the ‘cult-stand/standards’, should also be noted here. The ‘mangers’ and ‘cult-stands’ in front of the animal are fully integrated elements in the iconographic tradition of the Indus valley and must refer to some explicit ideological aspect the meaning of which now eludes us (for a discussion and suggestions see Vidale 2005: 153–155). While one of the two objects mentioned above is almost always present in front of the bulls on square Indus Valley seals, this feature is, as previously pointed out by Vidale (2005: 153), infrequent among other Gulf Type seals. Apart from the Group 1 seals, ‘mangers’ only appear with full certainty on seals 22 and 23 (Fig. 9/22, 23) and this feature is entirely absent from the almost 100 uninscribed seals of Gulf Type, with the sole

⁸ Franke-Vogt has further noted the general likeness of this composition with those found on some of the later Dilmun seals (1991: 99). The ‘whorls’ clearly bear some resemblance to Kjærøum’s motif group *Radial and whorl compositions* (1983: 14). The whorls on the later Dilmun Type seals have in turn been presented as one of several Syro-Anatolian influences that contributed to the shaping of this later iconographic tradition (Porada 1971: pl.10/8).

⁹ Seal no. 9 on Figure 9 is atypical in having nothing but a horizontal Indus inscription centred on the functional stamp.

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2.



23.



22.



16.



21.



26.



20.



25.



17.



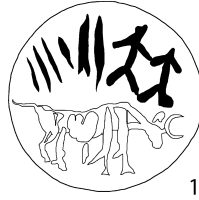
27.



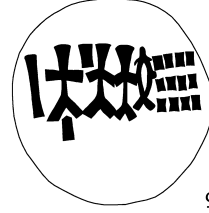
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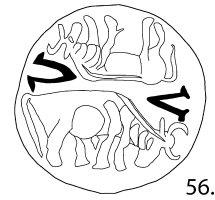
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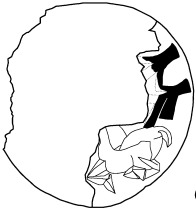
11.



9.



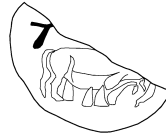
56.



6.



18.



19.



24.



13.



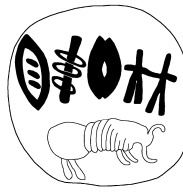
10.



12.



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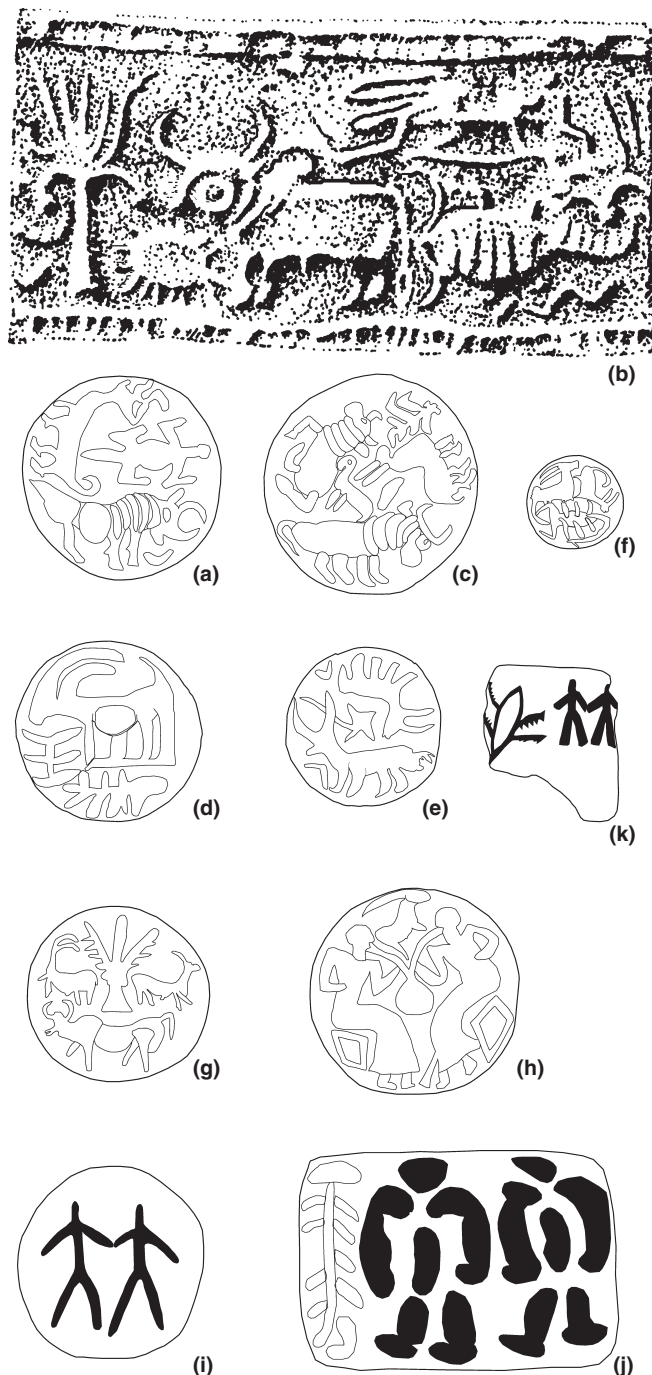


14.

Fig. 10.

a. An example of a seal with Indus-inspired bull without inscription (Table 1 no. 108) illustrated as an impression drawing (scale 1:1). Note the crescent-shaped 'manger' unparalleled in uninscribed seals; b. impression drawing of a cylinder seal from Ur with a humped bull and a 'bale of fodder' (not to scale) (Gadd 1932: pl. I/6), courtesy of Gregory L. Possehl; c. an example of a seal with Indus-inspired bull without inscription (Table 1 no. 57) illustrated as an impression drawing (scale 1:1); d–f. examples of Gulf Type seals from Bahrain in the 'local' style illustrated as impressions (scale 1:1); d (Table 1 no. 114) depicts two palm branches below a quadruped. e (Table 1 no. 32) depicts two quadrupeds and a pair of crescents while in the centre a 'comet' or 'shooting star' can be identified by its long tail; f (Table 1 no. 53) depicts a scorpion below a pair of quadrupeds; g is a unique example of a Gulf Type seal with Mesopotamian styled 'vulture' above a bull in profile; h is a Gulf Type seal depicting the classic 'two men drinking scene' (Table 1 no. 62), illustrated as an impression drawing (scale 1:1); i is from Kalba site K4 with a possible pseudo-'twins' sign (scale 1:1), after Cleuziou 2003: fig. 6/2; j is from Ra's al-Jinz RJ-2 also with a possible pseudo-'twins' sign (scale 1:1), after Cleuziou 2003: fig. 6/1.; k is a fragment of a 'cylinder seal' from Mohenjo-Daro with a 'twins' sign and another undistinguishable sign (scale 1:1), after Shah & Parpola 1991: 179, M-1370.

possible exception of a crescent-shaped object in front of the bull on seal 108 (Fig. 10/a).¹⁰ The mangers on seals 22 and 23 can be supplemented by an object that appears on seal 28 (Fig. 9/28) in the typical 'manger' position and which was originally interpreted as a flower, pomegranate or stylised tulip by Winkelmann (1999: 26). However, because of its characteristic location on the seal, the object is somewhat reminiscent of the 'mangers' on the classic seals — an identification, which admittedly on first impression is contradicted by its mystifying appearance. When consulting one of the Indus-styled cylinder seals published by Gadd (1932: pl. I/6) an interpretation of the object as some kind of feeding device does nonetheless seem justified (Fig. 10/b). This Ur seal, which is probably of early Isin-Larsa date, depicts an Indus-inspired bull



¹⁰ *Contra* Parpola who has previously stated that seals 19, 25, and 27 (as numbered in Table 1) also display 'mangers' (1994a: nos. 17, 34, and 41). Here I have chosen to disregard them because of the indistinct nature of the alleged mangers found on these seals. With regard to the identification of the object in seal a (Fig. 10/a) one should recall that crescents are one of the most frequent 'fill' object in Gulf and Dilmun seal compositions.

'feeding' from an object. This object is easily distinguishable from the various types of 'native' Indus 'mangers' and 'cult-stands' by its round form and emanating rays which could represent hay, making the object from which they branch out readily identifiable as a bale of fodder, as suggested by Gadd (1932: 196). In this light a similar interpretation

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seems warranted for the small round object on the atypical seal no. 28 (Fig. 9/28), from which three rays (of hay?) project towards the bull. Regardless of this second possible 'manger', the conclusion is that these iconic Indus symbols are generally disassociated from the iconographic tradition found on the Gulf Type seals of western provenance.

The seal now in the Ligabue collection (Fig. 9/28) is also distinguished from the sample by its inscription in Linear-Elamite (*contra* Proto-Elamite as stated in Vidale 2005: 151), which provides the seal with a strong link to western Iran. However, in spite of the somewhat inferior rendering of the bull, the marked heel (*calcaneus*) above the hoof, the raised and pointy tail, the anterior striations, the right-facing orientation and lowered head posture clearly illustrate that the seal was meant to 'mimic' the typical traits of Indus glyptic art. It can thus be argued to represent a hybrid of the western tradition of round seals with the standard composition 'bull in profile with Indus (Linear-Elamite) text'.

During Caspers (1972: 181) has stressed the wider significance of the curious fact that, while many types of bulls and wild animals appear on the square Indus valley seals — including buffalo, elephant, 'unicorn', rhinoceros, crocodile, hare and tiger — bison or short-horned bulls appear almost exclusively on the inscribed circular seals. Vidale goes on to suggest that the animals depicted in fact represent the wild gaur (*Bos gaurous gaurous*) and argues that the particular choice of this free-ranging, wild and powerful beast as an icon on the western seals should be explained in analogy to the dangerous and mobile lifestyle of the 'western traders' with whom he prefers to associate this class of objects (2005: 147 and 153).

From a glyptic and stylistic perspective it is important to emphasise that, although very similar, the bulls on the Gulf Type seals are rendered in ways that, with the exception of the fully native Indus-style bulls on seal no. 5 (Fig. 8/5) from Mohenjo-Daro and seal no. 23 (Fig. 9/23), are slightly different from those found in the Indus Valley tradition. In respect to seal 23 (Fig. 9/23), which additionally has a proper 'manger', this seal also stands out as a consequence of the particular Indus-like rendering of its heraldic bull. Thus, in the middle section of the bull one can observe a U-shaped 'Decke' feature which corresponds to

Franke-Vogt's variant 25 (1991: 114). The 'Decke' feature is not found on any other Gulf Type seals from the west, but it appears for example (as is typical for 'unicorn' seals in the Indus Valley) on the seal from Chanhudaro (Fig. 8/1).

Conversely, a bull of 'western' style is found on seal no. 2 (Fig. 9/2) from Mohenjo-Daro, which represents the only example where a seal with Indus provenance apparently embraces the 'western' bull-style variant, suggesting that this seal indeed represents a hybrid 'returnee'. In favour of this assumption one should recall that in the PCA, this particular seal was associated with Group 3, which is otherwise comprised of seals of Mesopotamian and Bahraini origin (see Fig. 4).

Generally, in terms of the morphology homogeneous seals of Group 3 are of particular interest because this category holds the majority of seals with inscriptions and because these, in respect to iconographic composition, are fairly standardised according to the Indus format (see Fig. 9). In all seals of Gulf Type with inscription the bull faces right (on the impression) and thus fully conforms to the principle of the Indus Valley iconography.¹¹ Having established these obvious similarities it is however also apparent that there are considerable differences in the manner in which the animals and individual signs are rendered. In this category there are a number of seals of various provenances, where the rendering of the bull is so unsophisticated that it leaves the impression that they came out of the workshops of seal cutters who were entirely inexperienced with this basic glyptic design. This is particularly the case with seals 11, 20 (not in PCA), 24 and 25 (Fig. 9/11, 20, 24 and 25), which were also outliers from Group 3.¹²

In some of the uninscribed seals the composition with a bull in profile on the 'baseline' is maintained while other seals have entirely abandoned this

¹¹ There are a few exceptions such as seal 6 in Figure 9 where the heraldic animal has been interpreted by Kjærsum as an antelope with a short tail (1994: 322), seal 27 in Figure 9 which depicts a peacock (?) and finally the — in many ways — unusual seal 24 which depicts a mating cow and bull facing left in impression.

¹² In seal 13 in Figure 10 the bull's head is carved with a compass drill, as is the convention with Kjærsum's earliest Dilmun Type style 1 (Kjærsum 1980).

standard grammar. Among the former there is a large number of seals where the bull is rendered in a mode identical to that found on the majority of seals with inscription (Fig. 10/a, c). That the 'illiterate' nature of these seals in fact represents a gradual movement away from the Indus tradition is further underscored by some of these bulls, which occasionally ('incorrectly') face left on the impression (e.g. Fig. 10/g). In the PCA this very frequent category of seals is found in both Groups 3 and 4 and is, in spite of the glyptic and iconographic influence from the Indus, without exception of Bahrain provenance.

In addition to the uninscribed seals with bulls on the baseline, one also finds another characteristic iconography predominantly on seals from Bahrain, in which (Fig. 10/d–f) caprids, ibexes, scorpions, human footprints, crescents and anthropomorphic figures dominate (Kjærøum 1994: fig. 1756/3–7, 9 and B–D). The Gulf Type seals belonging to the variant bearing this iconography are typically more crudely fashioned and the linear style in which they are rendered is less detailed than, and technically inferior to, the proper Indus and 'Indus-related' seals. In comparison, this inferior technical execution makes them stand out as distinctly 'local'. In the PCA this 'local' variant is chiefly found in association with Group 4, while the 'Indus-related' style found on both inscribed and uninscribed seals appears in Groups 2 and 3 and Group 3, respectively.

When considering the general source of the iconography found on the 'local' seals one is prompted to recall the Gulf Type seal (Table 1, no. 119) found in the 'Persian Gulf' room at Tepe Yahya (Lamberg-Karlovsky 1970: fig. 4/6), which displays some of the motifs and in particular, the characteristic linear style associated with this variant. However, apart from the Tepe Yahya seal and a couple of seals found in the outskirts of Damman (Table 1, nos. 118 and 120) and Failaka (Table 1, nos. 29 and 30) Gulf Type seals with the technically 'local' execution have not been recovered outside Bahrain. This is noteworthy because elements such as scorpions and human footprints were apparently widely used in the Near East, e.g. Susa (Amiet 1980: pl. 6/11), and were thus to some extent the result of external influence. A hybrid specimen linking all the above-mentioned iconographic traditions is found in seal no. 20 (Fig. 9/20) from the royal cemetery of Ur,

PG-401. This seal displays the classic composition of a (crude) bull in profile with 'Indus' text, but while one sign is recognisable as the plain 'man' sign and another is a variant of the 'crab' sign (see Pärpola, 1994b: fig. 5/1 no. 88), a scorpion and a human footprint typically found on the technically 'local' Bahraini seals are also mixed into this 'text'.

Finally, there are rare cases where the iconography of Gulf Type seals exhibits strong inspiration from the 'Mesopotamian' tradition, which in turn was to be deeply influential on the glyptic tradition of the succeeding seals of Dilmun Type (Kjærøum 1980). One exceptional Gulf Type seal thus depicts the classic Mesopotamian 'vulture' (Fig. 10/g) in a symmetrical manner that has strong parallels, for example with the glyptic art from the royal cemetery at Ur (see Woolley 1934: pl. 199, U.9513, U.12.350, U.9051, U.9618; pl. 207, U.11.863; pl. 211, U.17.812.). Below the 'vulture', which is flanked by two caprids, the seal features a bull in profile (facing left!) and thus clearly represents a hybrid of several traditions. Next to this seal is one that depicts the classic Mesopotamian 'two men drinking' scene (Fig. 10/h and Kjærøum 2007: fig. 3). As the 'two men drinking' scene became one of the most popular motifs after the transition to the Dilmun Type, one is left with the impression that these could be transitory seals characteristic of the latest Gulf Type. Given the relatively short duration of the Gulf Type this is not necessarily contested by the fact that both seal no. 62 (Fig. 10/h) and an atypical square stamp seal with a 'two men drinking' scene were found in graves in the earliest parts of the Karzakkan Cemetery (Laurson, in press: fig. 13/4 and 8).

In summary, it appears that although the 'bull in profile with Indus text' was an icon of the Indus Valley tradition, other regions such as Iran and Mesopotamia also contributed to the wide-ranging iconographic repertoire of the Gulf Type. Furthermore the relationship between the glyptic, iconographic and stylistic variation and the morphological groupings of the seals appears to support the archaeological validity of the structures observed.

A tentative stylistic sequence can thus be proposed: four of the five seals from the Sind province were clearly made by Indus craftsmen, while the smallest seal (Fig. 9/2) possibly represents a 'returnee' made in a western workshop. The alien 'whorl'

composition found on one of the Indus seals testifies to influence from the Near East and thus indirectly provides this anomalous class of circular lower Indus seals with yet another link which ties Group 1 to the 'west'.

The inscribed seals of Groups 2 and 3 appear to represent a primary 'western' Gulf Type that once it had emerged, was quickly isolated from the native Indus tradition. That this process was indeed of an abrupt nature is testified by the swift loss of the 'manger' and the fact that the occurrence of a bull with 'Decke' only occurs once in the seals of the western tradition (Fig. 9/23). Moreover as proper 'mangers' in the west are exclusively found in association with the two most deeply Indus-inspired seals from Mesopotamia and (possibly?) the seal with Linear-Elamite inscription, it seems that the cultural setting for this primary hybridisation is to be found outside the Indus Valley. The fact that the bulls found on the inscribed seals occasionally display a virtual collapse of artistic quality suggests that the transmission of knowledge/influence from schools of Indus seal makers at least in some milieus had ceased completely.

Because local iconography principally takes over the position formerly occupied by the inscription, stylistically the uninscribed seals that maintain the 'bull in profile' grammar can hardly be interpreted as anything other than a 'secondary' variant. Significantly, these relatively frequent seals have only been reported from Bahrain and thus represent the earliest variant of the Gulf Type to be geographically limited to a single area in any substantial numbers.

Also predominantly found in Bahrain is the stylistic variant which here is labelled the 'local' style because of the artistically primitive and technically unsophisticated linear renderings found on these seals. The iconography found on the 'local' seals possibly reflects traditional beliefs of the indigenous Dilmun population and draws on a symbolism that was shared widely throughout the Near East. At this point it is difficult to elaborate further on the relation between these 'local' style seals and other variants, other than to suggest that they may represent the products of autonomous and less specialised seal cutters. In conclusion to this tentative stylistic sequence one should finally advance the exceptional seals with Mesopotamian

iconography which testify to the fact that the Gulf tradition, although modestly, was also influenced from this direction.

The inscriptions/epigraphy

As stated in the introduction, the inscribed Gulf Type seals assume a vital position in the present attempt to explore the primary adoption and consequent emergence of indigenous sealing technology in the Dilmun culture.

The inscriptions in Indus characters, which at present are known from twenty-eight seals of Gulf Type have, since the seminal contribution by Gadd (1932), nourished linguistic and epigraphic discussions within the scholarly community (e.g. Hunter 1932; Parpola, Parpola & Brunswig 1977; Brunswig, Parpola & Potts 1983; Parpola 1994a; Vidale 2005). Common to most studies have been the attempts to match the observed sequences from the Near East with those known from the Indus Valley. In all attempts, the emphasis has been on evaluating whether the language used on the circular seals was different from or identical with that used for native Harappan inscriptions. Employing this method, Hunter (1932) was the first to reach the conclusion that the circular seals from Mesopotamia had unsupported sequences, whereas the four circular seals from Mohenjo-Daro (Fig. 8/3–5 and Fig. 9/2) and the classic square Indus seals recovered in Mesopotamia showed sequences well supported in the Indus Valley (1932: 469).

The more recent study (Parpola 1994a) of all Indus inscriptions found outside the Harappan sphere have shown that there is much variation as to whether or not support for the western sequences can be found in the Indus Valley. In order to evaluate whether these inscriptions were written in native Harappan or some foreign language, the semantic sequence of each inscription was compared with those recorded in the list of native Indus inscriptions. A. Parpola's study (1994a) includes Indus inscriptions from seals of prismatic, square, rectangular, circular and cylindrical shape as well as sealings and graffito on pottery. Some of the observed variation can be explained by the typological variation of the objects investigated and whether or not these represent obvious exports from the Indus region such as, e.g. pre-firing seal impression in Harappan pottery. Other cases are more

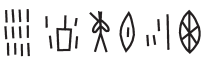
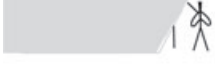
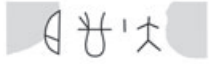
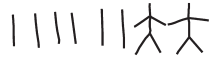
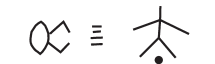
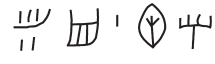
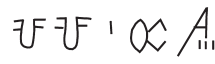
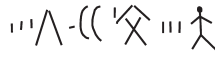
Group 1 (Indus Valley) 1.  3.  4.  5.  Group 2 (Iran) 14.  15.  28.  (linear-Elamite) PCA-outliers (Failaka) 12.  13.  (Bahrain) 10.  (Mesopotamia) 24. 	Group 3 (Bahrain) 6.  7.  8.  9.  11.  56.  Group 3 (Indus Valley) 2. 	Group 3 (Mesopotamia) 16.  17.  18.  19.  20.  21.  22.  23.  25.  26.  27. 
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Fig. 11. Inscriptions in Indus and Indus-related characters on Gulf Type seals as seen on the impression. The numbers refer to seal numbers in Table 1. Note the general abundance of ‘twins’ signs, especially at the beginning of the sequences.

ambiguous because inscriptions on objects of the same class such as the round seals produce contradictory results.

Aspects of the inscriptions found on the Gulf Type seals will be re-assessed here in their own right in full awareness that any conclusions on these matters

are tentative, given the unresolved issues surrounding the Indus script and the — in all likelihood — incomplete sample of textual syntaxes available from the Indus ‘motherland’.

For this purpose a comprehensive up-to-date list of the inscriptions has been compiled (Fig. 11), which, with the exception of the ‘new’ inscriptions, only diverge from the list in a minority of sequences previously published by A. Parpola (1994a: figs 1712–1719). The inscriptions have been written as they would have looked in impression, and the commonly accepted fact that the inscriptions were originally meant to be read from right to left should be noted.

The vantage point of this re-appraisal of the Gulf Type seals and of utmost relevance are the conclusions by Asko Parpola that, of the complete sequences found on Gulf Type inscriptions, only those from seals 14, 15 and 22 (Fig. 9/14, 15 and 22) are truly well paralleled in the corpus of Harappan inscriptions (1994a: nos. 22, 30 and 31). A. Parpola also considers that other more partially supported sequences may in fact be native Harappan, but here the evidence is less conclusive. The fact that Parpola identifies these three seals is highly interesting because this compliments what was observed earlier in this study. Seals 14 and 15 thus represent the sole examples of the Gulf Type with Indus inscriptions found in Iran. Together with the hybrid (Fig. 9/28) with Linear-Elamite text, these two morphologically similar seals (Group 2) were moreover demonstrated to be distinct from the major cluster of inscribed seals from Mesopotamia and Bahrain associated with Group 3 (see Figs 5 and 7). Consequently, it is interesting that the seal from Girsu (Fig. 9/22), which stands out from the rest of the sample by having a thin disc characteristic of the ‘Iranian’ seals (Fig. 7), also possesses a sequence supported in the Indus Valley. In the PCA analysis (Figs 5 and 7) seal 22, precisely because of its thin disc, was positioned in the periphery of Group 3, approaching the ‘Iranian’ seals in Group 2. This makes the fact that the Girsu seal is one out of only two seals of non-Indus provenance with a proper ‘manger’ appear even more significant. Here it should also be noted that in the PCA we also find seal no. 2 (Fig. 9/2) from Mohenjo-Daro (also with an Indus Valley-supported sequence) in a peripheral position in Group 3 approaching Groups 1 and 2, again point-

ing to a closer cultural and chronological relationship between the seals with supported sequences.

The correlation in the present study of the ‘independent’ variables of morphology and geographical provenance with Parpola’s Indus-supported sequence indicates that at least the seals from the Indus Valley and Iran (Groups 1 and 2) and seal no. 22 from Girsu were associated with communities of ‘Harappan language’ literates. Conversely, as seals with unsupported sequences cluster in Group 3 this adds renewed credibility to the hypotheses (Hunter 1932; Parpola, Parpola & Brunswig 1977; Parpola A 1994a; Vidale 2004: 265; 2005: 150) that a language other than that of the Harappan culture was employed on the Gulf Type seals found in Bahrain and Mesopotamia.

In our two ‘new’ inscribed seals lies evidence which lends added support to the notion that inscriptions on the seals which ‘circulated’ between South Mesopotamia and Bahrain were not written in the language(s) of the Harappans. This however requires a closer look at the hypothesis which surround the curious fact that the ‘man’ and related signs, especially the ‘twins’ sign, occur at abnormally high frequencies in the inscriptions on Gulf Type seals.

To the present writer’s knowledge research into this enigmatic subject was launched with a footnote by Parpola, Parpola and Brunswig in which they claimed that ‘the Indus inscriptions where the picture of “man” follows what can be presumed to be a god’s name in genitive case...’ could be compared with similar grammar in Sumerian (1977: 164 and n. 57). Even if unsubstantiated, Parpola, Parpola and Brunswig’s suggestion was the first attempt to tie the iconographic connotation of the ‘man’ sign with personal names. Later in his paper Asko Parpola (1994a) made a small, but important comment on the significant fact that whereas ‘twins’ signs (see Parpola A 1994b: fig. 5.1 no.16) begin inscriptions in a mere three texts in the Indus Valley, they do this in as many as four inscriptions in the Near East (1994a: 309). Parpola’s observations were later elaborated upon by Vidale who argued that the general abundance of variants of the ‘twins’ and ‘man’ related signs in the western seals may well testify to the presence of patronymic components in these inscriptions (Vidale 2004: 265; 2005: 156–157). Vidale has demonstrated the relative

high frequency of these signs in the inscriptions previously found to be non-Harappan by Parpola (Vidale 2005: fig. 5). In conclusion to these patterns Vidale ventures the suggestion that ‘...a correlate of my hypothesis is that the “man” and “twins” Indus signs, in the inscriptions from Failaka and Bahrain (and Ur?), might be interpreted as patronymic logograms, to be phonetically read in one or more (still unidentified) ancient Semitic languages’, and he goes on to predict that, ‘If in the future more inscriptions from Bahrain will be found (where the most substantial group was found), we might be able to test this idea by looking for ordered family sequences...’ (2005: 156).

It appears that Vidale’s prophecy has come at least partly true since we now stand in a position where the hypothesis that names (patronymic, personal or group affiliation) were communicated through the inscriptions can be further substantiated by the ‘new’ inscribed seals from Bahrain. In this respect it is first and foremost astonishing that the inscriptions found on the two ‘new’ seals (Fig. 11/9, 11) both exhibit ‘twins’ or much related signs (see Asko Parpola’s pers. com. at n. 2), making the hitherto observed abnormal frequency of these signs in Bahrain appear even more significant.

If attention is turned to all the inscriptions from Bahrain and Mesopotamia (Fig. 11 Group 3 and no. 24) one can generally observe a host of ‘twins’ signs and see that related signs also of anthropomorphic nature are abundant. Four inscriptions from Bahrain (Fig. 11/6, 7, 8 and 11) show that from this small island alone, there are now more inscriptions that start with a ‘twins’ sign than there are ‘twins’ signs occupying this position in all of the Indus Valley. At the same time several ‘twins’ appear deeper in the sequences of other inscriptions, which brings the total number of ‘twins’ signs in the twenty-seven Indus-inscribed Gulf Type seals to eight.

In this connection the most informative of the ‘new’ inscriptions is the pseudo-text found on seal no. 11. This is because even if the inscription is unambiguously of a pseudo-script nature the seal cutter apparently ‘remembered’ to add the ‘twins’ sign as a ‘prefix’. That the sign appears in its ‘typical’ position apparently suggests that even if the seal cutter obviously was an Indus script illiterate he/she had conscious knowledge of its connotation. One could explain the residual of precisely this sign in

several different, but mutually supportive, ways. Firstly, judging from the high frequency of inscription on seals in Group 3 (and Group 2) which start with the ‘twins’ sign, these could have easily passed before the eyes of our seal cutter when performing his profession, e.g. from seals fashioned by individuals more familiar with writing the hypothetical ‘Gulf lingo’ in Indus characters. Illiterate or not, the meaning originally conveyed by the inscriptions would hardly have escaped the attention of a specialised seal maker and if he in this way had become familiarised with ‘twins’ at the beginning of these ‘codes’, it would have been logical to mimic this in order to make his pseudo-inscription appear more credible to its intended audience. Secondly, if the two human figures composing the ‘twins’ in fact signify family lineage or a patronymic component, then the sign in semiotic terms possesses an integral or iconographic reference to its meaning (Bal & Bryson 1991 [Eng. translation 2001]; Preucel & Bauer 2001), making it equally simple to memorise and ‘decode’ for our ‘illiterate’ seal cutter. If this indeed was so, it would potentially also explain the particular prominence and disproportional size afforded to the ‘twins’ sign in the large and in every way abnormal seal from Ur, seal 24 (Fig. 9/24). In this seal the ‘twins’ sign claims visual superiority over the other ‘characters’ in the inscription as well as over the central ‘mating bulls’ motif on the ‘baseline’. Here it is also of particular interest that on a fragment of what appears to be a unique cylinder seal from Mohenjo-Daro (Fig. 10/k) one also finds a ‘twins’ sign in the inscription (I thank Dr Asko Parpola for drawing my attention to this seal). Although this combination of ‘western’ cylinder technology and a ‘twins’ sign is popular in the west we may have yet another indication that this ‘sign’ assumes a particular semantic position in the language employed by merchants operating in the western orbit.

There is evidence which suggests that a similar twin symbolism, perhaps originating in a ‘local’ connotation/ideogram reading of the ‘twins’ sign, may have held a central position in the ideology of the communities on the southern coast of the Gulf. This compatible symbolism is found on the Oman peninsula and has in another connection been labelled ‘symbol of alliance’ by Cleuziou (2003: 145 and figs 5–6). Cleuziou advanced the representations

of paired individuals including that on the southern entrance stone of the Umm an-Nar tomb Hili 1059 (c.2300–2100 BC), a bifacial steatite seal from a supposedly early second-millennium context at Kalba (Fig. 10/i) and a square stamp seal from Ra's al-Jinz RJ-2.1, dated to the Akkadian period (Fig. 10/j). In spite of the apparent differences in date it is tempting to connect at least the pseudo-‘twins’ sign in the Kalba seal to the tradition of ‘twins’ signs found as a prefix in the hypothetical ‘Gulf lingo’ inscriptions.

Even if Vidale’s ‘ordered family sequences’ are still a desideratum, one must share his optimism that further understanding of the Indus script may one day come from Bahrain. In the present author’s opinion the name hypothesis originally proffered by Parpola, Parpola and Brunswig (1977) is at any rate, if not confirmed, then substantially nourished by the new semantic context of the ‘twins’ sign.

From the famous account of the ‘Meluhha village’ (2062–2028 BC) known from third-dynasty tablets, one learns that most if not all its Meluhhan inhabitants bore Sumerian names (Parpola, Parpola & Brunswig 1977: 150). From this enclave of presumably Harappan descendants now known to be synonymous with the town of Guabba near Girsu (Vermaak 2008) we thus, probably because of their advanced stage of integration into Lagash society, come no closer to understanding what type of names was appropriated by Harappans and thus what names one in theory may expect to find in the inscriptions on Gulf Type seals. Conversely, if we turn to a text of Akkadian date dealing with the payment of ten silver shekels for breaking the tooth of a servant named Urur we learn the name of the perpetrator, ‘a man of Meluhha’ called Lú-sún-zi-da, the meaning of which matches our evidence from the Gulf Type seals amazingly well, as pointed out by Parpola, Parpola and Brunswig (1977: 161) and (Vidale 2004: 263). Although formally Sumerian the name of the Meluhhan in question ‘does not really make sense in the Mesopotamian cultural sphere...’ and literally translates ‘Man of the just buffalo cow’ (Parpola, Parpola & Brunswig 1977: 161). It is obviously not possible to extract a pattern from this single instance, but the explicit reference in the Meluhhan’s name to what must be a bovid deity corresponds remarkably well to the heraldic animal of the Gulf Type seals even if these are always

represented as bulls (Vidale 2004: 263–264). One could with a fair portion of goodwill see a similar ‘obscured’ bovid name in the ‘inscription’ on seal 56 (Figs 9 and 11).

The renowned square seal from Ur with a bull below a cuneiform inscription (Gadd 1932: no. 1) provides little resolution, as all of the many readings of the heavily worn inscription are somewhat inconclusive. A personal name is however allegedly linked to an unpublished but possibly Gulf-related seal without provenance from the Cabinet des Médailles, which shows a bull below a cuneiform inscription that, according to Glassner, contains a patronymic component in Sumerian (Vidale 2005: 152–153 citing Glassner 2002: 361 n. 215).

The inscription on seal 28 (Fig. 9/28) has three symbols in Linear-Elamite. The first sign read from left in the impression was originally suggested by Bork (1905) to represent a locative suffix indicative of the first person, which indicates that we could be dealing with a personal name. According to Winkelmann (citing Koch, pers. com.) the text should be read tash-shi-te or tash-shi-hu, which is supposedly a *proprium*/proper noun, perhaps of some deity (1999: 28 and n. 14).

In conclusion, the evidence from the twenty-nine inscriptions makes it probable that the language communicated through the inscriptions on the Gulf Type seals with only a few exceptions was of non-Harappan origin. The abnormal frequency of the ‘man’ and ‘twins’ sign supports the case that the Gulf Type seals of Group 3 in particular, found in Mesopotamia and Bahrain, communicated a specific standard message. The pseudo-inscription on seal no. 11 suggests that the prefixed ‘twins’ sign (which is the only rule-like regularity so far attested in this ‘western written language’) has an integral or iconographic reference to its meaning, making a proper noun the most reasonable tentative reading. This claim may well find support in the more or less contemporary use of ‘twin’ symbolism on the Oman peninsula, which in turn may represent a parallel local pseudo-‘translation’ of the original Indus ‘twins’ sign.

Chronology of the Gulf Type

There is generally poor evidence for dating the Gulf Type seals, but fortunately the little that is available

either comes from chronologically informative contexts or exhibits features that cast an important light on the genesis and evolution of the type.

The epigraphy, iconography and overall grammar of the motifs found on the seals leave little doubt that the origin of the Gulf Type on a general level must be sought in the context of the Harappan culture. This impression is amplified by the preference of stamp over cylinder technology in this class of objects and the classic 'Harappan'-styled pierced boss with a single groove perpendicular to its perforation (Bibby 1958).

In the Harappan culture the first prototypes of the classic square seals with early Indus script appear during the Kot Diji Phase or Early Harappan Period (2800–2600 BC) as testified by a terracotta sealing with an impression of a square seal with Indus characters and a square steatite seal with an elephant and possibly broken-off text found at Harappa (Kenoyer 2006: 15). The earliest seals with unambiguous text comes from Harappan Period 3A levels at Mound E dating from around 2600–2400 BC (Dales & Kenoyer 1993; Kenoyer 2006: 19), and from these early manifestations the seals seem to have rapidly developed into the mature square form that remained in use until c.1900 BC (Kenoyer 2006: 11).

In the Indus Valley and on the Indian subcontinent there are many indications that various round forms are introduced for stamp seals from the early second millennium BC onwards. This development is perhaps most clearly expressed by the tradition of the so-called Jhukar seal-bead amulets, c.1900–1700 BC, first known from Chanhudaro in Sind province (Mackay 1943: pl. XLIX and L). Piggott has argued for a central West Asian influence as responsible for the Jhukar seals (Piggott 1952: 226 cited by Miller 2008: 288), while Miller in her re-evaluation of the evidence from Chanhudaro has convincingly demonstrated that the Jhukar seals can equally be regarded as the products of an internal development that followed the major socio-economic re-organization of Harappan society (2008: 288). A comparable tradition has more recently been identified for the Ahar-Banas complex in Rajasthan (Shinde, Possehl & Ameri 2005). In the latter location one of the largest sites, Gilund, produced evidence of a flourishing tradition of circular stamp seals and pottery sealing tentatively dated from the late third to the

early second millennium (2005). Parallels have been drawn between the sealing tradition of the Ahar-Banas complex and the possible clay tokens found in several early second-millennium BC contexts in Bahrain, including the Barbar temples and the Early Dilmun settlement at Saar (Potts 2005: with references).

The existing chronology of the proper Indus seals does not allow for a precise dating of the four examples carved in traditional Indus Valley style from Mohenjo-Daro and Chanhudaro (Fig. 8) other than in the later parts of Harappan Phase 3 (c.2600–1900 BC). However, all the specimens from Mohenjo-Daro come from the upper levels (Marshall 1931, II: pls CII/k, l, m, CXII/383, CXIV/478; Mackay *et al.* 1937–38: pl. XCVI/500) and starting around 2000 BC a gradual, but not necessarily uni-linear, replacement of square-shaped seals for round forms has been observed for the region. Furthermore, the 'whorl' motif on the Chanhudaro seal (Fig. 8/4) appears in very similar form on the Dilmun Type seals from c.2000 BC. Given the combined evidence a tentative c.2200–2000 BC date is considered here.

Other, and perhaps more secure evidence for dating the variants of the Gulf Type seals with Indus text, comes from the context of those found by Woolley in PG 401 and PG 1847 in the Royal Cemetery of Ur (Gadd 1932; Mitchell 1986) whereas the information on the remaining seals from Mesopotamia offers only minor resolution.

Seal no. 20 (Fig. 9/20) was recovered in PG 401, which Nissen dates to c.2200–2000 BC ('Neu Sumेरische Zeit') (1966: 169). Seal no. 21 (Fig. 9/21), was found in the fill of the shaft leading to PG 1847. The stratigraphic relation between the seal and the establishment of PG 1847 is thus inconclusive, but among the excavator's different considerations he derived in conclusion that the seal was '...contemporary with and belonging to the grave.' (Woolley 1934: 192). Although Woolley originally dated shaft grave PG 1847 to the 'Second Dynasty' (1934: 192) it was later re-dated to the third dynasty of Ur by Nissen (1966: 106 and 191). With regard to the date of seal no. 21 (Fig. 9/21), Nissen advanced the important observation that the eastern corner of PG 1847's grave shaft was disturbed by the later construction of the 'Third Dynasty mausoleum': 'So erhalten wir durch die Stratigraphie eine Datierung der Schachtgräber in die Zeit von der späten akka-

dischen Zeit bis Amarsu'ens, dem 3. Herrscher der 3. Dynastie von Ur (2047–2038 BC), dessen Grabbau eine Ecke des PG 1847 zerstörte.' (1966: 106 [paragraph inserted]). Provided that the association between the 'Third Dynasty mausoleum' and the historic rulers of the third dynasty of Ur is correct, this would offer a date of the construction of PG 1847 *ante quem* the construction of the north-western section of the 'Royal' mausoleum, halfway through the reign of the third dynasty of Ur. Such a date is not contradicted by Pollock's analysis of pottery from the Ur cemetery, which demonstrates that Woolley's pottery Type 44, which at least in two instances occurs in PG 1847 (burial R) (Woolley 1934: 195, fig. 61) was used between the early Akkadian and post-Akkadian period (Pollock 1985: 138 and fig. 2). Gibson's analysis of the Umm al-Hafriyat and Nippur material has demonstrated that equivalent pottery marked an even more limited period spanning the late Akkadian to the early Ur III period (Gibson 1981: 79). The chronology of this particular vessel type is highly important because the same type of pottery (M11), with a single exception, appears exclusively in period Ib (c.2100–2050) at Qala'at al-Bahrain (Højlund 1994a: 105). Moreover, this Mesopotamian vessel type was also frequent in the so-called Early Type mounds in Bahrain, but its disappearance in the funerary assemblage coincides with the emergence of the vast mound cemeteries and primary appearance of Gulf Type seals in the burial mounds (Laursen, in press). In conclusion, it thus seems most plausible that both seals (Fig. 9/20, 21) date immediately prior to, or perhaps more likely very early in, the Ur III period.

Fundamental new information on the chronology of the Gulf Type seals with text has come in the form

of the seal from the western Iranian plateau (Fig. 9/28). It so happens that the association of this circular Gulf-related seal with 'bull in profile with text' and the Linear-Elamite inscription establishes a strong chronological link between the Gulf Type and the historic sequence of Mesopotamia. This fortunate situation arises from the fact that the use of Linear-Elamite appears to have been exclusively associated with the reign of Purzur-Insusinak, last king of the Awan dynasty.¹³ As Purzur-Insusinak, through Old Babylonian copies of Ur III royal inscriptions from Isin, is shown to be a contemporary of Ur-Namma, founder of the third dynasty of Ur (Potts 2008: 187 note 50), we can derive a date for the seal to around 2112–2095 BC corresponding to the reign of Ur-Namma in the Middle chronology.

The 'bale of fodder' (as opposed to a proper manger) on seal no. 28 was discussed earlier, but its presence is potentially also of chronological significance as it may reflect that this, and the two Mesopotamian seals with proper mangers (Fig. 9/22, 23), were produced at a time before the importance of this Indus symbol had, so to speak, been 'lost in translation'. A date to the earliest parts of the 'Gulf Type sequence' would also explain why these seals (Fig. 9/22, 23) have a much stronger 'Indus' touch to their bulls than on all the other western seals. This can, for instance, be observed in the 'Decke' on seal no. 23. This assumption is further supported by the 'native' Harappan sequence found on the Girsu seal because one can presuppose a logical order of acculturation where the Indus inscriptions, at least initially, were composed in 'native' Harappan, only later to be 'translated' (and consequently obscured). If this argument is accepted, the mangers would by implication provide us with a tenuous *post quem* of Ur-Namma (2112–2095 BC) for the vast majority of the Gulf Type seals that do not feature 'mangers'; but given the circumstances this remains speculation for the time being.

From Bahrain evidence for the chronology of the Gulf Type seals has come from various contexts that appear to complement each other well. These include Qala'at al-Bahrain, the Early Dilmun Settlement at Saar and the burial mounds.

From Qala'at al-Bahrain there are seven seals of Gulf Type all found in the period IIa levels dated to c.2050–2000 BC (one possibly in period Ib). Two of these have Indus inscriptions and five are

¹³ It is of interest to our case here that Potts has suggested that the centre of Awan was located in Luristan (where seal no. 15 was found) until Purzur-Insusinak at some point seized control of Susa and Elam (Potts 1999: 97–98; 2008). Until the death of his father, Purzur-Insusinak appears to have been governor of Susa where he made extensive constructions on the citadel. Most Linear-Elamite inscriptions date from his reign, for which reason he has been intimately associated with the use of this short-lived script in an unknown Elamite language. The invention of this independent Elamite script has been seen as an attempt by Purzur-Insusinak to strengthen the social cohesion of Awan by rejecting the use of Acadian from his court.

uninscribed, but unfortunately the stratigraphic evidence is not sufficient to establish whether there is a stylistic development through period IIa. The seals of Dilmun Type proper completely replace the Gulf Type from period IIb (c.2000–1900 BC) — a transition that is possibly marked by a single proto-Dilmun seal from the period IIa levels (Kjærøum 1994).

The Early Dilmun settlement at Saar is of equal importance to the chronology of the Gulf Type seals and in particular the transition to the later Dilmun seal tradition. At Saar this situation arises from the fact that only seven seals of Gulf Type were recovered, while seals of the Dilmun Type have occurred more abundantly ($n = 89$) (Crawford 2001). The London-Bahrain Archaeological Expedition has divided the stratigraphy at Saar into a number of Site Levels (Killick & Moon 2005), supplemented by some more or less corresponding Pottery Periods (Carter 2005). The earliest Pottery Period 1 is associated with some deeper levels (Site Level I) that have only been tested through minor soundings, which is why very little pottery and no glyptic material are available (Crawford 2001: 39; Carter 2005: 276). The pottery from this virtually unexplored Site Level 1/Pottery Period 1 corresponds to that from Qalāʾat al-Bahrain period IIa, and the excavators consequently expect that more extensive excavation of these deeper layers will produce seals of Gulf Type (Crawford 2001: 39). The Gulf Type seals that were recovered all came from Site Levels II and III, which roughly correspond to Qalāʾat al-Bahrain periods IIb and IIc, respectively. The excavators are confident that these Gulf Type seals are either intrusive from the late third-millennium deposits or, as suggested by Crawford, that 'their presence could be explained as that of survivals or heirlooms' (2001: 39).

The substantial negative evidence for Gulf Type seals from the Saar settlement thus confirms the observations from Qalāʾat al-Bahrain, and the conclusion is that the utilisation of Gulf Type seals in settlements on Bahrain Island was generally confined to Qalāʾat al-Bahrain period IIa (2050–2000 BC) after which they were more or less abruptly replaced by seals of the Dilmun Type in Qalāʾat al-Bahrain period IIb. Furthermore, since the Dilmun Type is securely linked to the historical sequence of Mesopotamia by the famous impression of a style I seal on a tablet dated to the tenth year in the reign of king

Gungunum of Larsa (1923 BC) (Hallo & Buchanan 1965; Kjærøum 1980), a launch date of the Dilmun Type around 2000 BC appears solidly founded. Given the combined evidence it would thus seem that one should consider the length of any transitory period where the Gulf and Dilmun Type seals coexisted to have been fairly brief.

The observations from the Early Dilmun settlements in Bahrain are complemented by evidence from the Early Dilmun burial mounds from where new evidence has recently been presented (Larsen, in press). This demonstrates that in Bahrain the Gulf Type seals appeared in the tombs exactly at the time when the burial praxis changed from scattered cairn-like mounds of Early Type to Late Type conical mounds concentrated in vast mound cemeteries (see Lowe 1986 for similar conclusions). Through an analysis of the 'horizontal' stratigraphy of the Karzakkan Cemetery based on changing patterns in burial mound architecture and pottery and seal distributions, it was additionally demonstrated that Gulf Type seals clustered in the largest of presumably numerous 'proto-cemeteries'. After the burial mounds had accumulated in this 'proto-cemetery' they became surrounded by later burials, many of which now featured seals of Dilmun Type. Two of the seals with Indus text (Fig. 9/8, 56) were found in the cluster of Gulf Type seals in this 'proto-cemetery'. The combination of inscribed and uninscribed Gulf Type seals observed in the Karzakkan Cemetery is analogous to that which could be observed in the period IIa deposits at Qalāʾat al-Bahrain, but unfortunately the data from the Karzakkan cemetery is not sufficient to elucidate further on the chronological relationship between these two Gulf Type variants.

From Failaka Island on the north-western fringe of the Dilmun sphere, the chronological evidence on the Gulf Type is chiefly negative. Here only two examples (exclusive of the Dilmun Type seal with pseudo-Indus script [Fig. 9/13]) have been reported — a number that is in striking contrast to the approximately 500 seals of Dilmun Type that have so far been unearthed from this small island. The Failaka evidence clearly strengthens the general assumption that the two seal types were chronologically separated. This is underscored by the unique Dilmun seal with unparalleled 'Indus' script (Fig. 9/13), which, with its bull's head/eye carved in the classic compass

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drill technique, moreover conforms to Kjærum's later Dilmun glyptic style I (1994). On the whole this seal definitely stands out as later than the Gulf Type seals but remains somewhat enigmatic both because of the alien/unparalleled signs in the inscription and the abnormally large diameter. This seal and, for that matter, also seal no. 12 from Failaka, may very well represent 'replicas' made from the models of heavily curated/heirloom seals with Indus inscription.

The Gulf Type-related seal from Tepe Yahya (Table 1, no. 119) was found in the 'Persian Gulf room' that is ascribed to phase IVB5 by Potts, who dates this phase to the last centuries of the third millennium BC (Potts 2001: 105). This position is, however, challenged by Lamberg-Karlovsky who argues for a date from 2400 to 2100 BC of phase IVB (Lamberg-Karlovsky 2001: 276). In respect of Lamberg-Karlovsky's objection to Potts's dating of phase IVB5, he proffers the 'Persian Gulf' seal as an argument of his higher chronology rather than the lower chronology suggested by Potts. In Lamberg-Karlovsky's argumentation it appears that he has misread the conclusions of Mitchell (1986) because he advances Mitchell's 1986 paper in a claim that the 'Persian Gulf' type is of pre-Akkadian date (Lamberg-Karlovsky 2001: 274), for which the evidence

certainly does not provide any reasonable support. Potts has subsequently presented more evidence for lowering the regional Kerman chronology including that of the Bampur sequence, based among others, on the black-on-grey canisters found in the well-dated tomb at Tell Abraq (Potts 2003).

At any rate, if one accepts that the 'Persian Gulf room' seal is a creation of the 'Gulf tradition', which I am inclined to do, its presence would certainly lower the date of Tepe Yahya phase IVB5 rather than raise it. The conclusion must be that until the local Kerman chronology is improved, limited faith can be vested in the prospect of the Tepe Yahya seal clarifying the chronology of the Gulf Type. However, with considerable reservation a date of the seal to c.2150–2050 BC is proposed here.

The fact that the seals of Group 4 are the ones which bear the closest morphological resemblance to Dilmun Type seals implies that they generally date in the later part of the Gulf Type sequence. This finds additional support in the high frequency in Group 4 of the seals with bull in profile without inscription as well as seals carved in the 'local' style. A chart with the basic chronological evidence discussed in this section is offered, together with selected historical 'landmark' information (Fig. 12).

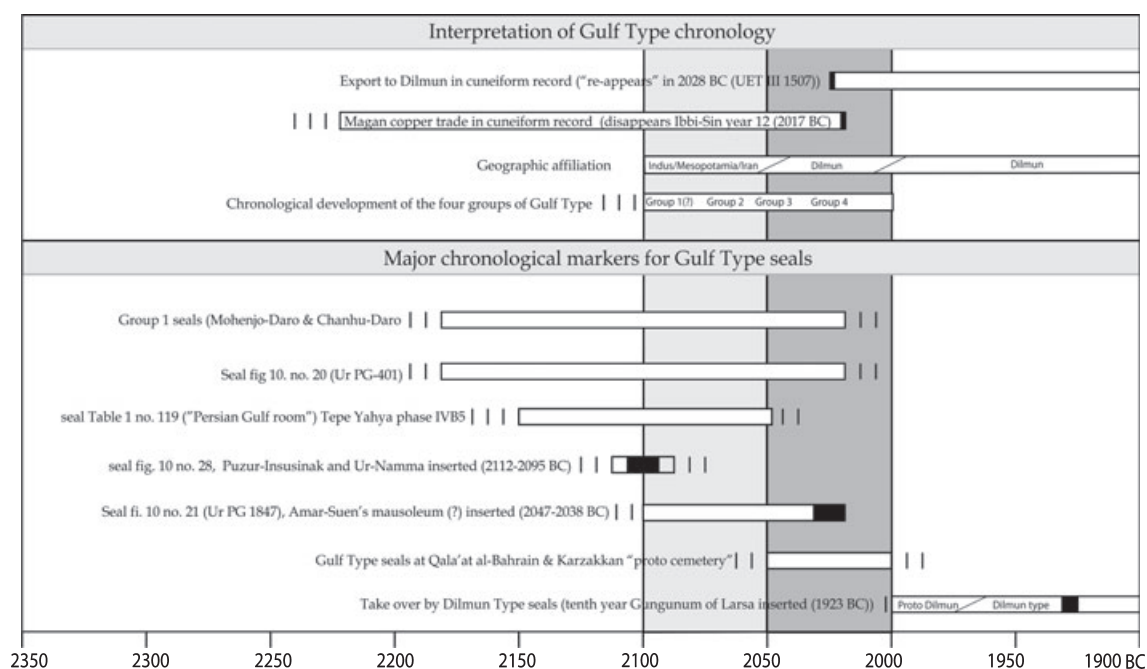


Fig. 12.

A Gulf Type chart showing important chronological markers and the interpretation presented in this article. Important textual/historical evidence has been included for comparison.

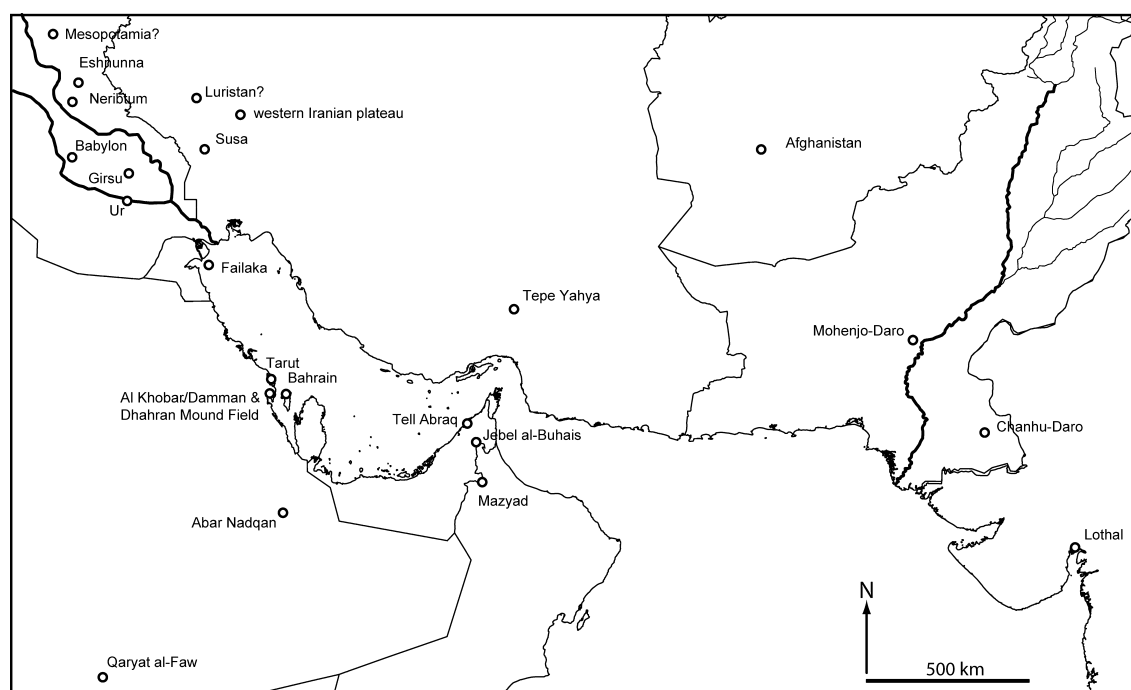


Fig. 13.

The distribution of Gulf and Dilmun Type seals demonstrates the geographical vastness of the underlying networks of exchanges.

Geographical distribution and discussion

Observing the accumulated distribution of all currently known circular seals of Gulf and Dilmun type one is struck by the vastness of its geographic range (Fig. 13). While some seals possibly travelled beyond the borders of the network of the Gulf trade in the hands of non-Dilmun agents, most seals probably testify to the actual degree of integration of Dilmun into an immense network of long-distance exchange. However, during the almost 500 years (c.2100–1650 BC) in which the circular stamp seals were in use, this dynamic network underwent a number of substantial changes.

In this study the focus has been on the emergence of sealing technology in Arabia and as will be evident, the geographical distributions of the major Gulf Type variants prove particularly instructive on this development.

At this point it can hardly be disputed that stamp-seal technology, either directly or indirectly, was transmitted from the Indus Valley, and it is consequently necessary first to consult the distribution of classic square Harappan seals in the 'west'. The number of actual Harappan sealing implements in the Near East and Arabia is limited,

but importantly concentrated in Mesopotamia. Discounting the western periphery of the Harappan sphere, a mere seven classic square seals have been found which correspond to those of the Indus Valley and of these, six come from Mesopotamia while one slightly atypical specimen of copper has been recovered in eastern Oman¹⁴ (Fig. 14 with references). Considering the amount of fieldwork carried out over the last fifty years on the Oman peninsula and on Bahrain, one should not expect this picture to change considerably. The occurrence of these seals in Mesopotamia has repeatedly been seen in connection with the presence of Meluhhan

¹⁴ As noted by A. Parpola (1994a: 315) the copper square seals from Ra's al-Hadd and Ur differ in material from the steatite conventionally used for seals in the Indus Valley, a fact which could also suggest that these particular seals were manufactured in some as yet unidentified production centre located between the two regions. A. Parpola suggests Oman as the origin on account of the material being copper. However, since steatite is also plentifully available here and because this part of Arabia did not develop a significant independent sealing technology, the exact origin of the rarer square seals of copper appears unresolved.

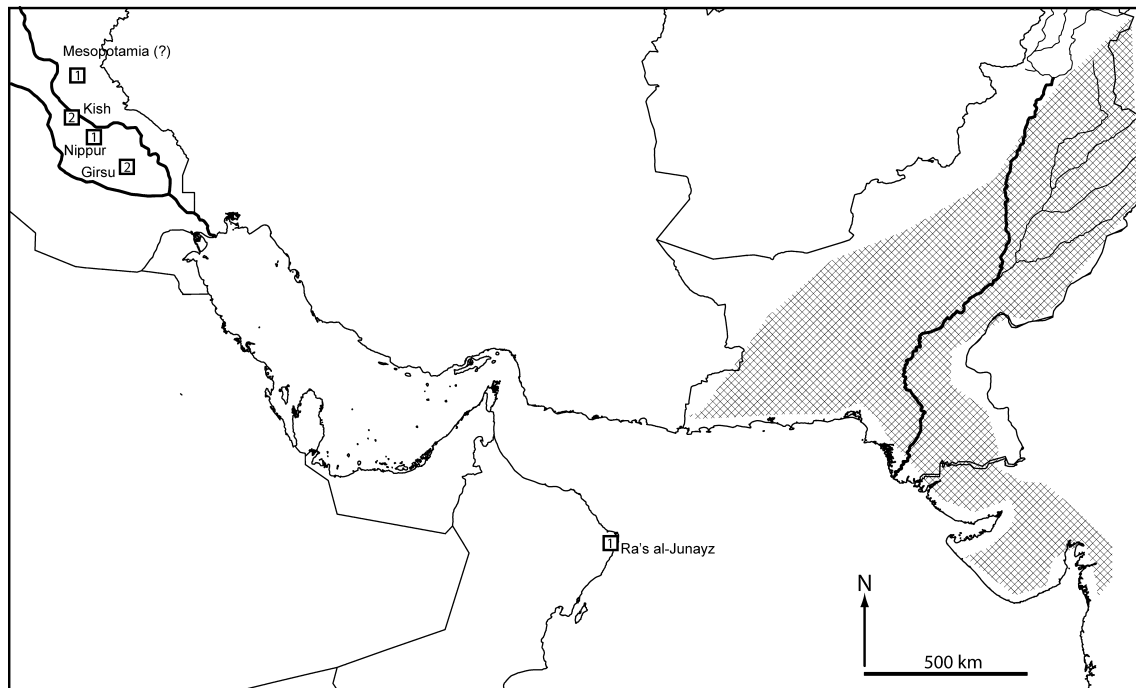


Fig. 14.

Classic square Indus seals. **Kish** (2): Langdon 1931: 593–593 and Mackay 1925: 679; **Nippur** (1): Gibson 1976: 26–28; **Girsu** (2): Thureau-Dangin 1925: 99 and Amiet 1988: 195 no.1; **Ra's al-Junayz** (1): Cleuziou & Tosi 1988: 12 and 21, fig. 18.1; 1990: 14 and 23, fig. 18; **Mesopotamia unspecified** (1): Brunswig, Parpola A & Potts 1983: 102–105 no.1 pl. I/1.

traders in Akkadian time as witnessed, for example, by Sargon's famous boast that the ships of Dilmun, Magan and Meluhha came to the docks of Akkad. The impact in Mesopotamia of material culture from the mature Harappan period including seals, and vice versa, is however remarkably faint.

The distribution of classic Harappan seals can thus be held against the distribution of Gulf Type seals with inscription, which occur in more substantial numbers (Fig. 15). The distribution of inscribed seals paints a relatively clear picture of a situation where this variant of the Gulf Type circulated between Bahrain Island and Mesopotamia (henceforth including Elam and Luristan for convenience).

As others have done before (e.g. Kjærum 1980; 1994; Mitchell 1986; Potts 1990), it is here argued that these seals represent the primary manifestations of the unique stamp seals that in gradually modified appearance were to stay in vogue for nearly half a millennium in the Dilmun culture. As has been discussed above, it is a distinct possibility that the seals from Mesopotamia and Iran with mangers

and/or Indus-supported sequences represent the very earliest Gulf Type seals. At the other end of the chronological spectrum of the inscribed seals, one finds the two seals from Failaka, which stand out in terms of morphology, iconography and glyptic style as late emulations.

Given the connection between the Linear-Elamite inscription and Purzur-Insusinak and the distinct possibility that his Awan centre shifted from Luristan to Susa as suggested by Potts (see n. 13), it appears highly probable that the hybrid seal no. 28 was manufactured in one of the two latter locations. Here a literate member of the 'Awan court' could have been inspired to manufacture the seal (hybrid) after affiliation or encounters with agents operating in the orbit between Mesopotamia and Bahrain. This suggestion is further collaborated by the fact that the only other two Gulf Type seals with inscription found in Iran derive precisely from Luristan and Khusistan (Susa).

Generally, the geography of the sites where inscribed Gulf Type seals have been unearthed is either characterised by a strong connection to the maritime network, be that directly or linked up by

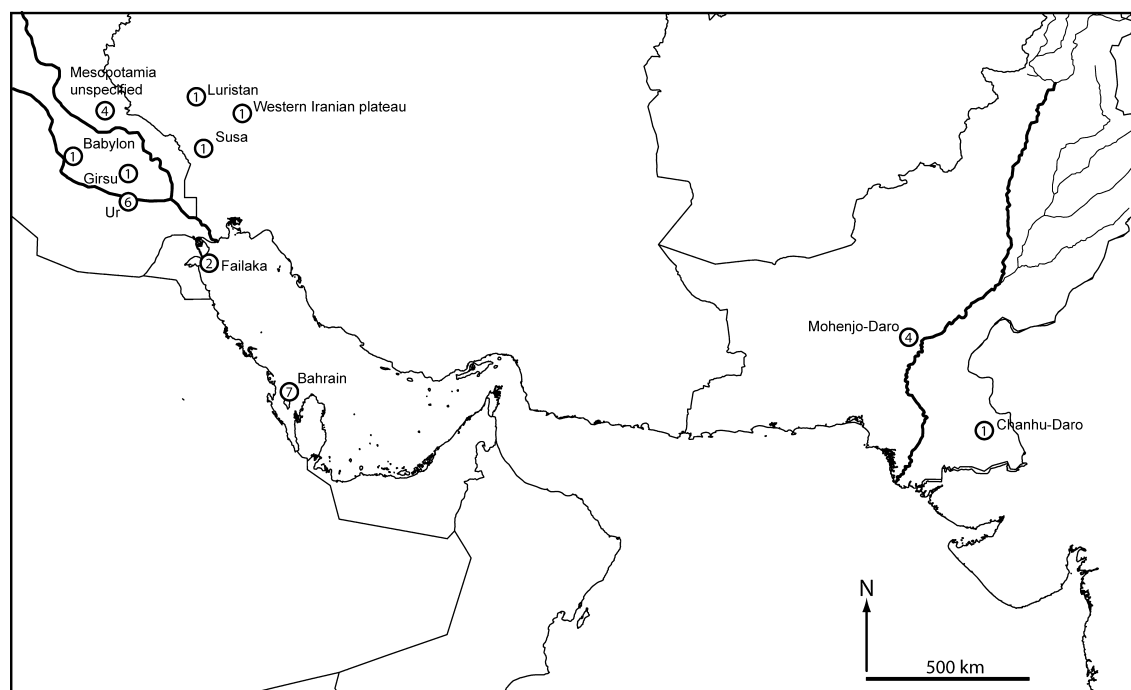


Fig. 15.

The geographical distribution of Gulf Type seals with inscriptions ($n = 28$ [27]): **Chanh-Daro** (1): Mackay 1943: pl. LI/23; **Mohenjo-Daro** (4): Marshall 1931: pl. CX/309, pl. CXII/383, pl. CXIV/478; Mackay *et al.* 1937–38: pl. XCVL/500); **Bahrain** (7): two in Kjærøum 1994: figs 1725–1726); one in Srivastava 1991: fig. 55; one in Al-Sindi 1999: no. 182; three in this paper nos. 10, 11 and 56; **Failaka** (2): Kjærøum 1983: no. 279 (technically Dilmun Type) and 319; **Ur** (6): Gadd 1932: pl. I/2–5 and pl. III/15–16; **Mesopotamia unspecified** (4): Gadd 1932: pl. III/18; Langdon 1932: p. 48; Buchanan 1981: nos. 1088–1089; **Girsu** (1): Sarzec & Heuzey 1884–1912: 321–322 and pl. 30.3a–b; **Susa** (1): Amiet 1972: pl. 153/1643); **Luristan** (1): Amiet 1973: pl. 23a–b); **Western Iranian plateau** (1) (Linear-Elamite inscription): Winkelmann 1999: Abb. 2.; **Babylon** (1): Collon 1994: 216 Babylon *contra* Gadd 1932 who at the time of his paper thought that ‘his’ seal no. 17 was unprovenanced).

ivers, or by compatible sites in the ‘lowland’ regions west of the Zagros Mountains. The vast ‘empty’ void between this cluster of seals in the west and the Harappan centres of the Indus Valley in the east must reflect a partially authentic situation, at least in archaeological terms. The dealings of the agents involved in this orbit left few or no traces in the intermediary regions of Iran or the Oman peninsula. This is significant because it shows that these seals from the onset circulated most intensely in a tight maritime orbit between Bahrain at the one end and major ‘Mesopotamian’ centres at the other. The morphological analysis and the stylistic examination suggest that one of the seals from Mohenjo-Daro (Fig. 9/2) should possibly be regarded as a product of the ‘western’ tradition. As such this seal perhaps indicates the easternmost boundary of this early ‘western’ orbit, which is not to be matched until later in the Isin-Larsa period, from whence a Dilmun Type seal discovered at the site of Lothal in Gujarat

(Rao 1963) is testimony of a compatible eastern range of the interaction network.

In respect of the four round seals from Mohenjo-Daro and Chanh-Daro (Fig. 8) that were shown to be products indigenous to the Harappan culture, two alternative cases seem to suggest themselves (if one excludes the western and eastern round seals as representing isolated developments). In a first scenario the alien round shape may have been adopted by seafaring Harappans after interaction with Gulf Type seal-using agents of the Mesopotamia-Bahrain orbit, who in turn had previously adopted this technology after contact with Indus Valley seafarers employing classic square seals. The evidence in favour of this argument is scarce but one could advance the western ‘whorl’ motif on one seal and possibly, as indication of contact, the western styled seal found at Mohenjo-Daro (Fig. 9/2).

In an alternative scenario the round form could have been the ‘trademark’ of some as yet undiscovered

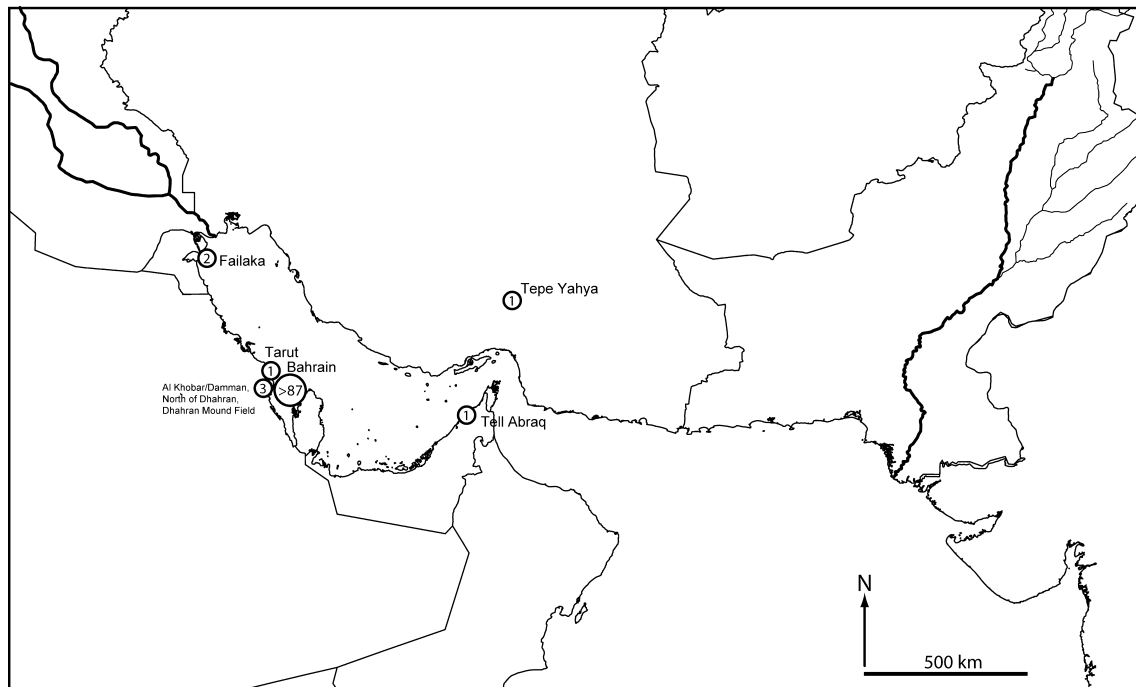


Fig. 16.

The geographical distribution of Gulf Type seals without inscriptions ($n = >95$): **Bahrain** (>87): seven in Crawford 2001: nos. 2622:05, 4197:03, 4139:01, 4300:01, 5506:05, 6581:02, L18:27:07; eight in Kjærøum 1994: fig. 1727–1734; fifteen in Kjærøum 2007: nos. 1–15; one in Srivastava 1991: fig. 55; one in During-Caspers 1977; one in Beyer 1989: no. 249 (*contra* Beyer not found at Qala'at al-Bahrain!); seventeen in this paper; four in Ibrahim 1982: pl. 61:2–3, pl. 60:3–4; twenty-nine in Al-Sindi 1999: nos. 2, 8, 65, 71, 86, 89, 98, 126, 129, 130, 133, 134, 136, 137, 138, 159, 160, 162, 163, 164, 165, 166, 174, 179, 181, 237, 284, 298, 310; three in Mughal 1983: pl. XIV/2, 4–5; one in McNicoll & Roaf 1975: pl. III/B–C; **Tepe Yahya** (1): Lamberg-Karlovsky 1970: fig. 4.6; **Tarut Island** (1): Zarins 1978: pl. 70/583; **Tell Abraq** (1): Potts 2000: 122; **Dhahran Mound field** (1): Presinger 1983: 1161, fig. 186/1; **Al Khobar/Damman** (1): Barger 1969: 139–140; **North of Dhahran** (1): Golding 1974: 19–31; **Failaka** (2): Kjærøum 1983: nos. 294–295.

ered urban Harappan centre deeply involved in the westbound maritime trade as originally suggested by both Gadd (1932: 204–205) and Mackay (1948: 343). If the four indigenous Harappan seals were in fact produced at some 'phantom' Indus Valley site, this would certainly explain both their relative and absolute infrequency at Mohenjo-Daro and Chanhudaro as well as their relatively standardised dimensions. However, more than fifty years after Gadd and Mackay arrived at their hesitant conclusions, any Indus Valley site where these round steatite seals outnumber, match or even approximate the number of their square counterparts still remains to be discovered. When the numerous cities and hundreds of unexcavated towns and villages which today are known to have made up the fabric of Indus society (Wright 2009) are taken into consideration, it nonetheless cannot be excluded that some trade-oriented settlement of considerable size, responsible

for the production of round steatite seals of the distinct variant (Group 1) found in Chanhudaro and Mohenjo-Daro, is awaiting future detection.

Conversely, when focus is turned towards the distribution of all the variants of Gulf Type seals without inscriptions a geographic pattern emerges, which contrasts that of the inscribed seals (Fig. 16). It is first and foremost striking that no seal from this category has been reported from Mesopotamia.¹⁵ The seals from Tepe Yahya, Tell Abraq and Failaka can all be explained as anomalies in one way or another and this leaves us with one massive cluster

¹⁵ This absence can only in part be explained by their general inconspicuousness in comparison with the seals with Indus characters, and their consequent higher chance of being overlooked in older excavation collections and in the antique market, and must accordingly reflect a 'real' trend.



Fig. 17.

Twelve of the Gulf Type seals mentioned in the text which have not previously been published. Numbers refer to seal numbers in Table 1. Note that photographs of seals 49, 78, 80, 81 and 115 were either not available or the seal was too fragmented for documentation.

of uninscribed Gulf Type seals on Bahrain Island and adjacent Damman Dome.

How should the contrasting distributions of inscribed and uninscribed seals be explained? In this paper it has thus far been argued that the disappearance of inscriptions represents a gradual chronological development, but considering the contrasting distributions this may need to be nuanced. One could speculate that many of the uninscribed seals from Bahrain, in addition to functioning as seals primarily served as visual emblems, which conveyed their owner's more or less symbolic affiliation with the new and prestigious identity of the seafaring merchant. This leaves us with the distinct possibility that some of the geographical variation between uninscribed and inscribed seals should be attributed to, respectively, a 'local' and an 'international' sealing system encompassed within the Gulf Type tradition. If this were so, it would be only natural to see the 'local' styled seals from Bahrain as a fairly isolated phenomenon closely linked to the forging of these new types of identities among the broader Dilmun population. At this point in time it can safely be expected that the broader population in Dilmun did not participate in the orbit of long-distance trade and consequently only utilised seals domestically for endemic sealing purposes, and perhaps equally importantly, as emblems of cultural and individual identity.

In contrast, inscribed (and undoubtedly some uninscribed) seals with bull in profile were apparently reserved for the sealing praxis exercised by a minority of long-distance seafaring merchants, as testified by their appearance in both Mesopotamia and Bahrain (and in one possible case, Mohenjodaro). In the morphological analysis it was demonstrated that collectively the inscribed seals found in Bahrain and Mesopotamia formed one homogeneous group (Group 3) and while the seals from Iran (Group 2) have slightly contrasting dimensions, the evidence overall suggests that the inscribed seals originated in one and the same 'population'. However, one can hardly ignore the epigraphic, stylistic and iconographic features, which point to an older date for some of the seals from Mesopotamia and Iran.

But what can be said at present about the locale of the earliest Gulf Type production and the cultural affiliation of the merchants who first utilised them? Judging from the close link in the glyptic art to the Indus Valley tradition and the adoption of the round

'Harappan' boss with a single groove, there can be no doubt that Harappans or acculturated Harappans schooled in the craft of seal-making, were somehow involved in the cultural transmission. The glyptic styles are profoundly influenced by the Indus Valley tradition, whereas Mesopotamian iconography has only been observed on three seals of Gulf Type all of which were found in Bahrain.

These 'native' Harappans must at least periodically have resided in Mesopotamia and Bahrain (?), undoubtedly in connection with seasonal trade expeditions. We can also safely assume that at some point (c.2100 BC) a relatively small group of Harappans broke away from the Indus Valley 'establishment'. They developed their own type of stamp seal, which was consciously designed to be distinct from both the classic square Indus seal and the Mesopotamian cylinder seal.

The large contemporary community of Harappan descendants, which settled in 'the Meluhha village' Guabba, is known to have used cylinder seals with cuneiform inscriptions. These Meluhhans were in a very detectable way highly integrated into the administrative praxis of this Mesopotamian state. That the cylinder technology and Mesopotamian iconography were rejected for the Gulf Type seal thus appears to be a significant clue as to the non-Mesopotamian cultural focus of these seal-making entrepreneurs.

As previously suggested by Parpola (1994a), the round stamp may find its origin in the Iranian communities utilising this technology at the beginning of 2100 BC and perhaps the evidence presented in this paper on the earliest seal of Groups 2 and 3 strengthens this assumption.

Consequently, it is here suggested that the setting of the cultural transmission that lead to the first Gulf Type seals must be sought somewhere in a dynamic orbit connecting southern Mesopotamia, Iran and Dilmun, and that from the very beginning the agents responsible enjoyed stronger relations with Dilmun than to any centres (known to date) in their old Indus 'motherland' or Mesopotamia.

As discussed above, the glyptic and linguistic evidence suggests that a small handful of the inscribed seals found in 'Mesopotamia' are approximately half a century older (c.2100–2050 BC) than those so far reported from Bahrain. The first Gulf Type seals may have been produced in western Iran

but the absence of these — the very earliest Gulf Type seals — in the burial mounds of Bahrain does not exclude the presence of seal-making agents on Bahrain around 2100 BC. It could very well be that the traders who first utilised this autonomous seal type still followed the ‘die hard’ conventions of their ancestral (Harappan) burial customs, which excluded seals in their personal funerary assemblage, or simply that we have not found their graves.¹⁶

Be that as it may, the picture changes from around 2050 BC when Gulf Type seals with Indus text appear as personal grave-goods in fully Dilmunite mound interments, simultaneously with the emergence of the vast mound cemeteries on Bahrain (Højlund 2007; Laursen, in press). This links the widespread adoption of sealing technology with other major socio-economic changes in Dilmun and simultaneously the first Gulf Type seals also appear on Dilmunite settlements. The previously intense interaction with the Umm an-Nar communities (Magan) becomes almost invisible in the ceramic material in Dilmun (Laursen 2009 with references).

It would be reasonable to assume that the appearance of inscribed seals (and Gulf Type seals generally) in the graves is both the result of a gradual integration and consequent settling of ‘second’-generation Gulf Type seal-making ‘Harappans’ in Dilmun and the adaptation of the technology by the indigenous Dilmunite population.

The lack of other Harappan cultural traits in the material assemblage can be interpreted as the product of a process where the ‘Harappans’ rapidly assimilated local customs after marriage into indig-

enous Dilmunite families. For these breakaway ‘Harappans’ strategic marriages would have been an obvious way to solidify trade alliances and for the Dilmunite elite the perfect tool to acquire first-hand knowledge of the administrative technologies of the ‘Harappans’.

Production of Gulf Type seals from this period is well attested in Bahrain from the excavations at Qala’at al-Bahrain where a workshop with remains from Gulf Type seal production was found during the 1958 season (Glob 1959; Kjærøum 1994: 338, fig. 1753; Højlund 1994b: 394), and in the form of a discarded Gulf Type pre-form picked up between the Umm es-Sujur ‘well-temple’ and the Diraz settlement (During Caspers 1977).

The evidence from the western inscriptions in Indus characters suggests that the Indus script was rapidly adjusted to accommodate writing in a language other than that of the Harappans, as testified by the unsupported sequences and prefixed ‘twins’ signs. This development is best attested in the epigraphic material from Bahrain and would not have taken this ‘primitive’ form among acculturated Harappans situated in the highly literate communities of southern Mesopotamia.

Both the mention in the cuneiform texts of a Dilmun weight standard equivalent to that of the Harappans (Bibby 1971; Roaf 1982; Menderos & Lamberg-Karlovsky 2001) and the archaeological evidence from Bahrain (Bibby 1971: 345–353; Højlund 1994c: 395) demonstrate that coeval with the first seals and writing, Dilmun adopted its earliest weight technology from the Indus Valley.

The distribution of seals found in Mesopotamia suggests that, at this end of the orbit, the agents were set up in major centres of trade as part of a system that constituted a predecessor to the more governmentally sanctioned institutions which emerged later, as indicated by the institutionalised Dilmun seal proper.

The evidence appears to show that the agents operating in the orbit between Mesopotamia and Bahrain after only a few generations of Gulf Type seal production (c.2100–2050 BC) were attracted to the Dilmun polity from where they henceforth launched their trading expeditions (e.g. from commercial centres at Tarut and Qala’at al-Bahrain). If and when these Dilmunites operated/resided in Mesopotamia (Luristan, Khusistan and the Sind province [?]) this

¹⁶ Seals with Indus inscription have been found in at least five burial mounds on Bahrain Island and in two cases in Ur. These instances are in marked contrast to the fact that seals are virtually never found in association with burials in the Harappan culture (Kenoyer 2006: 17). In the Early Dilmun graves in Bahrain the inscribed seals are used as markers of personal identity, as are all other variants of the Gulf Type, the total number of which is now approaching 100. Judging from *in situ* positions near the neck of the buried individual in some graves from Bahrain, the seals appear to have been worn in a string around the neck. In the two graves from Ur the association with buried individuals is more ambiguous and here it cannot be excluded that Gulf Type seals represent foreign objects incorporated in the funerary assemblage more for their exotic prestige than strictly as personal effects.

was chiefly as participants in organisations perhaps compatible with the Assyrian Kārum.

Conclusion

The conclusion is an attempt to answer the questions put forward in the introduction to this study: by what route did this sealing technology spread? Who were the agents instrumental in its transmission? When did it happen and what role did it play in the emergence of social complexity in Early Dilmun society?

The innovative group of risk-taking entrepreneurs that were instrumental in transmitting Indus Valley sealing, writing and weight technology into Dilmun culture must at first have been composed of break-away Harappans (c.2100 BC), followed by a combination of Dilmunite and acculturated Harappans merchants (c.2050 BC) attracted by the emerging social elite to the rising centre of trade on Bahrain. The sealing technology was adapted in a relatively open and experimental environment, as evidenced by the presence of hybrid forms such as the prism-shaped seals and Gulf-styled cylinder seals. The crudely made 'local' styled seals on Bahrain show that seal production in the earlier phases was not monopolised by a central authority or exclusively maintained in the hands of specialists. Conversely, with the emergence of the Dilmun Type seal around 2000 BC the stamp seals became heavily institutionalised, as testified by the standard three grooves and four dots-in-circle 'brand' on the reverse.

The transformation of the Indus script into a 'western' grammar as testified by the prefix 'twins' on the seals suggests that the process occurred in relative isolation from the Indus Valley centres. This also seems to be the explanation for the almost instant 'loss' of the 'mangers'/'cult-stands' and 'Decken' in the 'bull in profile' composition. Seen together with the obvious de-selection of the classic square shape and the explicit use of the short-horned bull as a heraldic animal, the translated 'Gulf edition' of Harappan sealing culture appears as a hybrid seal designed to convey a strong message of autonomy. Thus, in establishing the 'Gulf Type' the

parties involved on the one hand borrowed technology and symbolism from a well-established sealing tradition, while on the other made a substantial effort to insure sufficient distance from the original tradition. It thus seems that the adaptation of an indigenous sealing technology (and other administrative technologies) may have been an integrated component in the endeavour of the indigenous Dilmun elite to challenge the monopoly of Magan on trade with Mesopotamia. One can speculate that Dilmun's adaptation of well-tested systems for sealing, writing and weighing did not provide a vital organisational edge that in the long run proved to be a decisive factor behind Dilmun's favourable economic position in the Isin-Larsa period.

Acknowledgements

This paper was written in the course of a PhD project based at Aarhus University and Moesgaard Museum. I would like to thank warmly the Bahrain Minister of Culture and Information, H.E. Shaikha Mai bint Mohammed Al Khalifa, for her generous support of the project.

I would also like to express my cordial thanks to the following people and institutions who have kindly supplied information on, or access to, the seals used for this study: Asko Parpola, University of Helsinki; Pierre Lombard, CNRS Lyon; Sidney Babcock, The Pierpont Morgan Library, New York; Mustafa Salman and Khalid Al-Sindi, Bahrain National Museum; Nabil Al-Shaikh, Dammam Regional Museum (KSA); Richard Zettler, University of Pennsylvania Museum; St John Simpson, The British Museum; Ulla Kasten, Yale Babylonian Collection; Dan Potts, University of Sydney; Poul Kjerum and Flemming Højlund, Moesgaard Museum; Nicole Chevalier, Marielle Pic and Béatrice André-Salvini, Département des Antiquités Orientales, Musée du Louvre.

I would also like to thank a number of people for their valuable assistance in various matters without which this paper could not have been written: Jens Andresen, Robert Carter, Flemming Højlund, Gregg Jamison, Mads Dengsø Jessen, Kasper Lambert Johansen, Asko Parpola, Gregory Possehl and Rita P. Wright. I also sincerely thank the two anonymous reviewers for their sharp and constructive comments on this paper, which were of great help.

Finally, I would like to acknowledge my debt to the pioneer of Gulf seal research, the late Poul Kjerum who sadly passed away while this manuscript was being prepared. I am grateful to him for his encouragement, help and many stimulating conversations about stamp seals in recent years.

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